

Tariff Shocks, Household Exposure, and Aggregate Demand*

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Abstract

Tariffs affect aggregate demand through the joint distribution of household import exposure and spending responses. I derive a path-weighted sufficient statistic that decomposes the direct purchasing-power effect into average exposure and between- and within-group covariance terms. A stylized small-open-economy HANK model with non-homothetic baskets illustrates the mechanism. Under the benchmark tariff process, sorting amplifies the fixed-basket statistic by 7%, mainly across expenditure groups. In general equilibrium, the expenditure gradient drives the heterogeneous-basket output gap, which nearly vanishes when basket composition is fixed. Imported-input tariffs account for most of the activity contraction, while final-goods tariffs generate unequal cost-of-living effects.

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1 Introduction

What is a tariff shock? In macroeconomic models, the standard answer is a cost shock. A tariff raises import prices, increases firms' marginal costs, pushes up inflation, and creates a trade-off between price stability and economic activity. With imported intermediate inputs, this logic is immediate: the tariff enters production costs and shifts the Phillips curve upward (Werning et al., 2025). In richer production networks, foreign price shocks propagate through input-output linkages, trade imbalances, and sectoral exposure to foreign inputs (Aguilar et al., 2026).

This view is correct but incomplete. A tariff is also a shock to household budgets. Even before the rest of the economy adjusts, higher prices for imported consumption goods mean that the same income buys less. The size of that loss depends on what a household buys, while its effect on spending depends on how easily the household can absorb it. Put differently, two households facing the same tariff may therefore experience different increases in living costs and adjust their spending by different amounts.

This observation separates two questions. The first is distributional: which households bear the larger direct cost? The second is macroeconomic: which households cut spending when that cost arrives? A tariff is more contractionary when the largest losses fall on households that respond strongly to changes in liquid resources. It is less contractionary when the most exposed households can smooth the loss. Unequal incidence, by itself, need not imply a large aggregate-demand effect.

I make this argument precise by deriving a sufficient statistic for the tariff's direct effect on household spending. Because a tariff shock may last for several periods, the statistic follows households' spending responses to the full path of purchasing-power losses rather than relying only on the impact. It separates the effect implied by average exposure from the additional effect created when exposure and spending responses are systematically aligned. The decomposition also shows what standard incidence data can reveal. Group averages by income, expenditure, liquidity, or region describe broad exposure gradients, but they do not reveal whether the most exposed households within each group are also the most responsive.

I then ask how much these distinctions matter once prices, production, and policy adjust. To do so, I embed the mechanism in a stylized small-open-economy HANK model and compare three economies. In the first, every household has the same consumption basket. In the second, import exposure varies systematically with expenditure. In the third, households with similar expenditure may still have different baskets. These comparisons isolate which forms of household heterogeneity matter for aggregate adjustment and which matter mainly for the distribution of tariff costs.

Three findings summarize the quantitative analysis.

First, who is exposed matters for direct spending, although the amplification is moderate. Under the benchmark tariff process, the alignment between exposure and spending responses makes the direct effect 7% larger than the effect implied by average exposure alone. Most of this alignment comes from differences across expenditure groups: lower-expenditure households are more exposed to imported final goods and tend to respond more strongly because they hold less liquid wealth.

Second, the expenditure gradient is the exposure margin that matters for aggregate activity. Allowing exposure to vary with expenditure deepens the initial output contraction, whereas adding basket differences among otherwise similar households produces almost no further aggregate change. In the model, households with necessity-like import demand switch away from imported goods less aggressively and redirect less spending toward domestic production.

Third, household exposure matters more for the distribution of tariff costs than for the size of the aggregate recession. Lower-expenditure households face larger increases in their cost of living and have less capacity to smooth the loss. Differences in baskets among similar households add little to aggregate output, but they widen the range of cost-of-living changes within expenditure groups.

The main message is therefore simple. A tariff can be strongly unequal without generating a large additional aggregate contraction through household heterogeneity. For aggregate demand, distribution matters when the households bearing the losses are also those most likely to reduce spending. The same distinction can help organize other relative-price shocks, including exchange-rate movements, energy-price shocks, and changes in consumption taxes.

Related literature. This paper connects four strands of literature. The first studies the household incidence of trade shocks through consumption prices. A central insight is that changes in trade prices do not affect all households equally because households buy different baskets. [Porto \(2006\)](#) and [Nicita \(2009\)](#) develop early survey-based approaches that combine consumption-price effects with income and factor-market channels. [Fajgelbaum and Khandelwal \(2016\)](#) show that non-homothetic expenditure patterns make the gains from trade unequal across the income distribution. [Cravino and Levchenko \(2017\)](#) document heterogeneous cost-of-living effects following large devaluations, arising both across broad consumption categories and within categories.

More directly related to trade protection, [Artuc et al. \(2021\)](#) quantify household exposure to agricultural tariffs, while [Jaccard \(2022\)](#) emphasizes the role of household-level substitution in measuring tariff incidence. [Borusyak and Jaravel \(2024\)](#) show that trade shocks can generate substantial horizontal inequality: households with similar income can face very different price changes because they consume different goods.

This paper builds on that consumer-side view, but shifts the object from exposure alone to exposure interacted with spending responses. Unequal exposure determines who bears the direct cost of a tariff. Its aggregate demand effect additionally depends on whether the exposed households have high or low marginal propensities to spend. The paper therefore moves from a distribution of cost-of-living losses to a joint distribution of losses and spending responses.

The second literature studies heterogeneous marginal propensities to consume and aggregate transmission. Empirical work documents large and persistent differences in household spending responses. [Parker et al. \(2013\)](#) estimate sizeable consumption responses to the 2008 U.S. stimulus payments, while [Jappelli and Pistaferri \(2014\)](#) review the broader evidence on MPC heterogeneity. [Kaplan and Violante \(2014\)](#) show that a model with liquid and illiquid assets can generate large responses to temporary transfers. [Kaplan et al. \(2014\)](#) document the importance of wealthy hand-

to-mouth households, who own illiquid wealth but hold little liquid wealth and therefore respond strongly to transitory resource changes.

HANK models embed these spending responses in general equilibrium. [Auclert \(2019\)](#) decomposes monetary transmission into intertemporal substitution and redistribution across households with different MPCs, balance sheets, and income exposures. [Kaplan et al. \(2018\)](#) show that changes in labor income, profits, and asset returns can dominate the direct intertemporal-substitution channel. [Auclert et al. \(2024\)](#) develop the intertemporal Keynesian cross, in which matrices of intertemporal MPCs summarize the propagation of household income changes into aggregate demand.

The sufficient statistic in this paper applies the same logic to a different impulse. A tariff raises the expenditure required to purchase a household’s consumption basket. The resulting aggregate demand effect depends on how those purchasing-power losses are allocated across the distribution of intertemporal MPCs. The relevant object is not the average MPC or average exposure separately, but their expenditure-weighted interaction.

The third literature asks when household heterogeneity matters for aggregate fluctuations. The approximate-aggregation result of [Krusell and Smith \(1998\)](#) shows that large household-level differences need not imply large aggregate effects. In New Keynesian settings, the answer depends on the shock and policy environment: heterogeneity matters when high-MPC households are differentially exposed, but aggregate responses can remain close across RANK, TANK, and HANK when that covariance is weak or policy offsets it ([Kaplan et al., 2018](#); [Bilbiie, 2025](#); [Debortoli and Galí, 2025](#)). [Bilbiie et al. \(2025\)](#) provide direct empirical support for this joint-exposure view: tax-and-transfer insurance substantially reduces the aggregate-demand amplification implied by heterogeneous labor income exposure. This paper applies the same aggregation question to household exposure to tariffs.

The fourth literature studies tariffs and foreign-price shocks in open-economy macroeconomic models. Evidence from recent U.S. tariffs documents substantial pass-through into duty-inclusive import prices and important effects on prices, trade flows, and welfare ([Amiti et al., 2020](#)). [Erceg et al. \(2018\)](#) show that the short-run effects of trade policy depend on nominal rigidities, expenditure switching, and the monetary response. [Lindé and Pescatori \(2019\)](#) revisit Lerner symmetry in a quantitative New Keynesian open economy. [Barattieri et al. \(2021\)](#) show that protectionism behaves like an adverse supply shock, raising inflation while reducing economic activity.

Open-economy HANK models add household real-income and multiplier channels to the standard expenditure-switching mechanism. [Auclert et al. \(2021b\)](#) show that exchange-rate shocks operate through expenditure switching, heterogeneous real-income effects, and aggregate demand feedback. Their use of non-homothetic consumption baskets is particularly relevant here: international price changes have stronger real-income effects when imported goods represent a larger share of the expenditure of high-MPC households.

Recent tariff models emphasize the distinction between final and intermediate imports. Tariffs on consumption goods raise household living costs and induce expenditure switching. Tariffs on imported inputs raise production costs and reduce domestic activity. [Auclert et al. \(2025\)](#) study the

short-run effects of tariffs on GDP and the trade balance in an open economy with intermediate-input trade, while [Auray et al. \(2025\)](#) emphasize how nominal rigidities and the monetary regime shape the macroeconomic response to trade conflict. This paper adds a distributional household block to that composite-shock view.

Roadmap. The rest of the paper is organized as follows: Section 2 isolates the direct purchasing-power channel and derives a sufficient statistic for the partial-equilibrium demand effect of a tariff. Section 3 embeds the mechanism in a heterogeneous-agent small open economy. Section 4 studies the quantitative version of the model. Section 5 describes the main findings. Section 6 concludes.

2 What is a tariff for a household?

Locally, a real resource loss. To show this, I isolate the direct cost-of-living channel of tariffs. Throughout this section, initial expenditure baskets are predetermined¹, non-tariff prices and incomes are held fixed, and fiscal transfers and tariff-revenue recycling remain fixed along the entire partial-equilibrium path. Therefore, the object is the local purchasing-power loss generated by a tariff-induced increase in household cost-of-living indices, and how this loss maps into current aggregate demand.

Figure 1 places this object within the broader anatomy of a tariff shock. Tariffs can trigger many adjustment margins, but the partial-equilibrium exercise focuses on the demand branch: prices rise, households are exposed through their consumption baskets, cost of living increases, and spending responds. The key question is whether the households most exposed to the shock are also those with stronger spending responses.

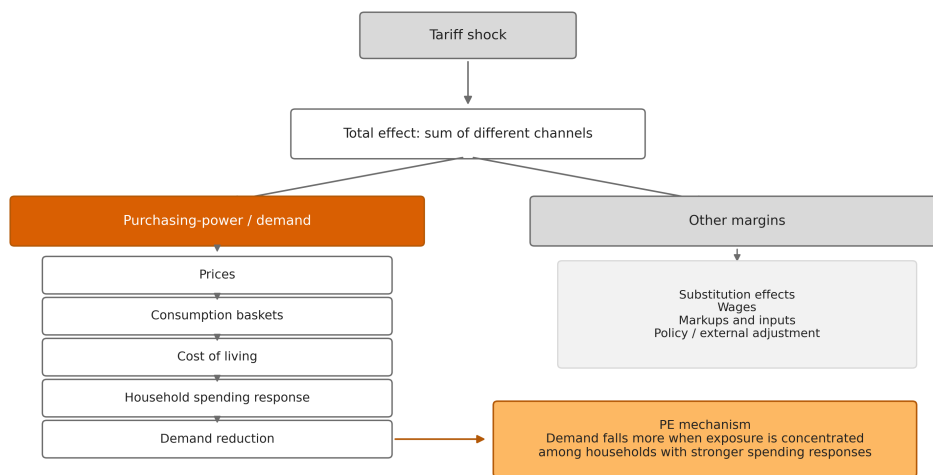


Figure 1: Anatomy of a tariff shock.

¹Predetermined baskets are used only as reference baskets for the local purchasing-power calculation. They need not coincide with the baskets households choose at later dates once substitution and general-equilibrium adjustment are allowed.

Household exposure. Consider a continuum of households indexed by $i \in \mathcal{I}$, with distribution μ . Household i has initial real consumption expenditure $c_i > 0$, and aggregate expenditure satisfies $C \equiv \int_{\mathcal{I}} c_i d\mu(i) \in (0, \infty)$. Define the expenditure-weighted expectation as

$$E^C[x_i] = \int_{\mathcal{I}} x_i \frac{c_i}{C} d\mu(i). \quad (1)$$

Households consume goods indexed by $g \in \mathcal{G}$. Let ω_{ig} be household i 's initial expenditure share on good g , with $\omega_{ig} \geq 0$ and $\sum_g \omega_{ig} = 1$. A local innovation $d\tau_0$ to the log tariff wedge induces

$$d \log p_{g,t} = \theta_g q_t^\tau d\tau_0. \quad (2)$$

The normalized path q^τ is deterministic, known at date 0, common across goods, and satisfies $q_0^\tau = 1$. Under full pass-through on tariffed imported goods,² θ_g is the relevant tariffed import intensity.

Household i 's tariff exposure is

$$\alpha_i^\tau = \sum_{g \in \mathcal{G}} \omega_{ig} \theta_g. \quad (3)$$

Assume that α_i^τ is measurable and bounded. Using the initial basket as the reference basket, its Divisia cost-of-living change is

$$d \log P_{i,s} = \alpha_i^\tau q_s^\tau d\tau_0. \quad (4)$$

Normalizing $P_{i,ss} = 1$, the equivalent liquid-resource loss is

$$dy_{i,s}^{eq} = -\alpha_i^\tau c_i q_s^\tau d\tau_0. \quad (5)$$

This is a local reference-basket valuation. In other words, the tariff is comparable to an income loss only for this local calculation; it is not globally equivalent to an income shock, nor does the argument require baskets to remain fixed after substitution and general-equilibrium adjustment.

Demand incidence. The purchasing-power loss affects current demand through households' dated spending responses. Let

$$M_{i,0s} = \left. \frac{\partial c_{i,0}}{\partial y_{i,s}} \right|_{ss} \quad (6)$$

denote household i 's date-0 spending response to a one-unit liquid-resource transfer at date s . The tariff-relevant intertemporal marginal propensity to consume is

$$\Psi_i(q^\tau) = \sum_{s=0}^{\infty} M_{i,0s} q_s^\tau. \quad (7)$$

²Full pass-through simplifies the interpretation of θ_g but is not required. More generally, θ_g collects the first-order consumer-price response to the primitive tariff wedge.

Under the boundedness and integrability conditions in Appendix A, household responses can be aggregated across households and time.

Applying the dated responses to the resource losses gives

$$dc_{i,0}^{PP} = -\Psi_i(q^\tau)\alpha_i^\tau c_i d\tau_0. \quad (8)$$

Proposition 1 (Demand sufficient statistic). *For a local tariff innovation $d\tau_0$, holding fixed households' initial reference baskets and expenditure bases and abstracting from general-equilibrium feedbacks, define*

$$s_\tau(q^\tau) = E^C[\Psi_i(q^\tau)\alpha_i^\tau]. \quad (9)$$

The direct aggregate spending response is

$$\frac{dC_0^{PP}}{C} = -s_\tau(q^\tau)d\tau_0. \quad (10)$$

Appendix A provides the formal proof. Defining $\bar{\Psi}(q^\tau) = E^C[\Psi_i(q^\tau)]$ and $\bar{\alpha}^\tau = E^C[\alpha_i^\tau]$ gives

$$s_\tau(q^\tau) = \bar{\Psi}(q^\tau)\bar{\alpha}^\tau + \text{Cov}^C(\Psi_i(q^\tau), \alpha_i^\tau). \quad (11)$$

The first term is the product-of-marginals benchmark; the second captures sorting between exposure and intertemporal spending responses.

Corollary 1 (Demand orthogonality). *For a nonzero local tariff innovation and given $\bar{\Psi}(q^\tau)$, average tariff exposure is sufficient for the direct aggregate spending response if and only if the covariance term in (11) is zero. For a positive tariff innovation under the nonnegative-response conditions in Appendix A, a positive covariance makes the spending decline larger, while a negative covariance makes it smaller.*

Vertical and horizontal sorting. The trade-incidence literature distinguishes exposure differences across socioeconomic groups from residual differences within them (Fajgelbaum and Khandewal, 2016; Cravino and Levchenko, 2017; Borusyak and Jaravel, 2024). Exposure dispersion determines unequal cost-of-living incidence, but aggregate demand amplification depends on whether exposure is aligned with spending responses across or within observable groups. I call these covariance components vertical and horizontal sorting.

Let $b(i) \in \mathcal{B}$ be a measurable finite or countable partition, with positive expenditure-weighted group shares $\pi_b^C \equiv \int_{\{i: b(i)=b\}} (c_i/C) d\mu(i) > 0$, and let $E_b^C[x_i] \equiv E^C[x_i \mid b(i) = b]$ denote the corresponding conditional expectation. Write $\bar{\Psi}_b(q^\tau) \equiv E_b^C[\Psi_i(q^\tau)]$ and $\bar{\alpha}_b^\tau \equiv E_b^C[\alpha_i^\tau]$ for the group means.

The law of total covariance splits sorting into a vertical component $V_\tau(q^\tau)$, the covariance between group-average exposure and spending responses, and a horizontal component $H_\tau(q^\tau)$, the average residual covariance within groups. Appendix A gives the formal definitions and derivation.

Both components depend on the partition. Vertical sorting captures alignment between group

means; horizontal sorting captures residual alignment within groups. In particular, H_τ is not a structural parameter and need not equal the effect of introducing permanent basket types in the quantitative model.

Proposition 2 (Vertical and horizontal sorting). *For a local tariff innovation, holding fixed the partial-equilibrium objects specified above, the sufficient statistic is*

$$s_\tau(q^\tau) = \bar{\Psi}(q^\tau)\bar{\alpha}^\tau + V_\tau(q^\tau) + H_\tau(q^\tau). \quad (12)$$

Together with equation (10), this characterizes the direct aggregate spending response. For $d\tau_0 \neq 0$ and given $\bar{\Psi}(q^\tau)$, aggregate average exposure is sufficient if and only if $V_\tau(q^\tau) + H_\tau(q^\tau) = 0$; group shares and group means are sufficient if and only if $H_\tau(q^\tau) = 0$.

Corollary 2 (Group data may not be enough). *Suppose two economies face the same tariff path and have, for every group b , the same expenditure-weighted group share π_b^C , average tariff exposure $\bar{\alpha}_b^\tau$, and average spending response $\bar{\Psi}_b(q^\tau)$. If the economies differ in $H_\tau(q^\tau)$, they have different normalized direct aggregate spending responses, dC_0^{PP}/C , to the same nonzero local tariff innovation.*

Group shares and means determine the average-exposure benchmark and vertical sorting, but not the within-group covariance. Hence identical group-level gradients need not imply identical aggregate demand effects.

3 The HANK model

Does the household-incidence mechanism survive once prices, incomes, and policy are allowed to adjust? To answer this question, I embed the mechanism in a small-open-economy HANK model. Household exposure to imported consumption goods has two structural sources: Stone–Geary demand creates an expenditure gradient in import shares, while permanent basket types create residual dispersion among households with similar expenditure. These sources need not coincide exactly with the between- and within-group terms of the sufficient statistic, which depend on how households are grouped in the data.

The aggregate structure follows the open-economy HANK framework of [Auclert et al. \(2021b\)](#) and the quantitative small-open-economy model of [Druedahl et al. \(2024\)](#). Its purpose here is narrower than providing a general theory of tariffs: the model traces how the joint distribution of import exposure and spending propensities shapes household incidence, aggregate expenditure, and the equilibrium response to a common tariff shock.

Households and permanent heterogeneity. Households differ along four dimensions: liquid assets a , idiosyncratic labor productivity e , a permanent discount-factor type β , and a permanent consumption-basket type ξ . Their individual state is therefore $z_{i,t} = (a_{i,t-1}, e_{i,t}, \beta_i, \xi_i)$, with productivity following a finite-state Markov chain with transition matrix Π_e . The two permanent types

play separate roles. Discount-factor heterogeneity helps generate the dispersion in liquid wealth and marginal propensities to consume, whereas ξ shifts import demand without directly changing earnings risk, patience, or access to financial markets.

In the baseline, the joint mass over β and ξ factorizes into the product of its marginals. Thus the model does not impose that impatient households have more import-intensive tastes. This assumption does not make β independent of realized exposure: patience affects saving and resources, resources affect expenditure, and Stone–Geary demand maps expenditure into import shares. The exposure–MPC relationship can therefore emerge endogenously even though the permanent types are independent ex ante.

At the household level, aggregate hours N_t and the real wage w_t are taken as given, while individual labor income is scaled by productivity $e_{i,t}$. Their equilibrium determination is described in the wage-setting block below. This timing leaves households with a tractable one-asset consumption–saving problem without suppressing the aggregate labor margin.

All real quantities are expressed in units of the domestic final good. Let $p_{M,t}^C$ denote the tariff-inclusive relative price of imported consumption goods. Household i then solves

$$V_t(a_-, e; \beta, \xi) = \max_{c, a'} u(c) - v(N_t) + \beta \mathbb{E}_t V_{t+1}(a', e'; \beta, \xi), \quad (13)$$

$$\text{s.t.} \quad \zeta_t + \mathcal{P}_{\xi,t} c + a' = (1 + r_t^a) a_- + (1 - \tau_t^\ell) w_t e N_t + T_{i,t}, \quad a' \geq \underline{a}. \quad (14)$$

Here r_t^a is the realized return on the liquid claim, τ_t^ℓ is the labor-income tax rate, and $T_{i,t}$ denotes transfers. The remaining two terms, ζ_t and $\mathcal{P}_{\xi,t}$, summarize the subsistence expenditure requirement and the price of consumption above subsistence; the next block derives both from intratemporal demand.

Incomplete markets make spending responses differ with liquid wealth, earnings risk, the borrowing constraint, and patience. Non-homothetic baskets then determine how strongly the tariff loads on each of those households.

Non-homothetic consumption baskets. A standard homothetic CES aggregator can generate heterogeneous MPCs, but at given relative prices it assigns the same import share to all households with the same preference weight. Consequently, differences in resources do not translate into differences in tariff exposure, and every household faces the same proportional purchasing-power loss. Because that restriction removes the mechanism of interest, I instead use a Stone–Geary CES aggregator. Household i combines domestic and imported final goods, $c_{H,i,t}$ and $c_{M,i,t}$, according to

$$c_{i,t} = \left[(1 - \omega_{\xi_i})^{1/\eta_C} (c_{H,i,t} - \bar{c}_H)^{\frac{\eta_C - 1}{\eta_C}} + \omega_{\xi_i}^{1/\eta_C} (c_{M,i,t} - \bar{c}_M)^{\frac{\eta_C - 1}{\eta_C}} \right]^{\frac{\eta_C}{\eta_C - 1}}. \quad (15)$$

Here η_C governs substitution between domestic and imported final goods, $\bar{c}_H$ and $\bar{c}_M$ are common subsistence requirements, and ω_{ξ_i} is the import weight applied to consumption above subsistence. The common subsistence component makes import exposure vary with total expenditure, whereas variation in ω_ξ creates residual basket dispersion among households with similar expenditure. The

two sources of heterogeneity are therefore distinct by construction.

Cost minimization gives $x_{i,t} = \zeta_t + \mathcal{P}_{\xi,t} c_{i,t}$, where the subsistence cost is $\zeta_t = \bar{c}_H + p_{M,t}^C \bar{c}_M$ and the supernumerary price index is $\mathcal{P}_{\xi,t} = [(1 - \omega_\xi) + \omega_\xi (p_{M,t}^C)^{1-\eta_C}]^{1/(1-\eta_C)}$. The resulting import expenditure share is

$$s_{M,i,t} = \frac{p_{M,t}^C \bar{c}_M + \vartheta_{\xi,t} \mathcal{P}_{\xi,t} c_{i,t}}{\zeta_t + \mathcal{P}_{\xi,t} c_{i,t}}, \quad (16)$$

where the supernumerary import expenditure share is $\vartheta_{\xi,t} = \omega_\xi (p_{M,t}^C / \mathcal{P}_{\xi,t})^{1-\eta_C}$. At fixed prices and basket type,

$$\frac{\partial s_{M,i,t}}{\partial \log x_{i,t}} = \vartheta_{\xi,t} - s_{M,i,t}. \quad (17)$$

When imported goods are necessity-like, $s_{M,i,t} > \vartheta_{\xi,t}$, equation (17) is negative: the import share falls as expenditure rises. Low-expenditure households then face the larger increase in their cost of living. Because those same households tend to hold fewer liquid assets, remain closer to the borrowing constraint, and display higher MPCs, Stone–Geary demand can align exposure with spending responses even when basket type and patience are independent.

Permanent variation in ω_ξ adds a different form of heterogeneity. It creates unequal exposure among households with similar resources and hence widens the distribution of incidence. Its aggregate effect, however, depends on whether this residual exposure is systematically associated with spending responses; dispersion alone is not enough.

Having specified household demand, I now introduce the two tariff-inclusive prices that connect households to the production and fiscal blocks.

Tariffs and import prices. Let $q_t \equiv E_t P_t^* / P_{H,t}$ denote the foreign price level in units of the domestic final good. Under producer-currency pricing and full pass-through, the tariff-inclusive prices of imported consumption goods and imported inputs are

$$p_{M,t}^C = (1 + \tau_t^C) q_t, \quad p_{M,t}^I = (1 + \tau_t^I) q_t. \quad (18)$$

The two wedges separate the household and production margins. A tariff on final imports raises the cost of the consumption basket, whereas a tariff on imported inputs raises firms' marginal cost; the common experiment studied below activates both. In either case, foreign suppliers receive the border price q_t . The difference between that price and the tariff-inclusive price paid domestically is fiscal revenue, not an additional payment abroad.

With the household price already embedded in the budget problem, the next step is to describe how the input price enters production and, through firms' decisions, domestic income.

Production. The production side distinguishes domestic value added from gross output. Labor first produces value added, $X_t = Z_t N_t$. A consolidated gross-output sector then combines the domestic input $X_{H,t}$ with imported intermediates $M_{I,t}$:

$$Y_t = \Gamma_Y \left[(1 - \alpha_I)^{1/\nu_I} X_{H,t}^{\frac{\nu_I-1}{\nu_I}} + \alpha_I^{1/\nu_I} M_{I,t}^{\frac{\nu_I-1}{\nu_I}} \right]^{\frac{\nu_I}{\nu_I-1}}. \quad (19)$$

The elasticity ν_I controls substitution between the two inputs. The fixed scale term Γ_Y reconciles this CES representation with stationary gross-output accounting and, because it is constant, disappears from the first-order dynamics. Imported intermediates affect costs and gross output but do not enter domestic value added directly. Hence a rise in $p_{M,t}^I$ induces substitution toward the domestic input and changes labor demand, with the strength of that response governed by ν_I .

Gross-output producers charge a flexible-price markup μ_p over the composite unit cost and distribute profits, net of the fixed operating cost F , to the domestic mutual fund. Holding F fixed in units of the domestic final good along the transition reconciles stationary markups, dividends, and household financial wealth. The resulting final good is absorbed by households, the government, and foreign buyers; export demand has constant elasticity η_X with respect to q_t , while foreign absorption and foreign-currency prices are exogenous.

This block determines labor demand and profits. Closing the income side of the model now requires wages, aggregate hours, and the monetary response.

Wage setting and monetary policy. The household problem took w_t and N_t as given. In equilibrium, a labor union chooses aggregate hours, allocates those hours across households, and resets nominal wages with a Calvo probability. Staggered adjustment leads to the wage Phillips curve

$$\pi_t^W = \bar{\beta} \mathbb{E}_t \pi_{t+1}^W + \kappa_W \hat{\mu}_t^W, \quad (20)$$

where $\hat{\mu}_t^W$ is the wage-markup gap.

Because an additional hour produces different labor income across productivity states, the wage-setting condition weights marginal utility by productivity. Product prices, by contrast, are fully flexible. This asymmetry keeps the nominal block parsimonious and directs attention to household real-income transmission rather than optimal inflation stabilization. Aggregate CPI inflation is built from compensated household expenditure indices, so it reflects heterogeneous stationary baskets instead of imposing a representative household basket.

In first-order log deviations, monetary policy follows the inertial Taylor rule

$$\hat{i}_t = \rho_i \hat{i}_{t-1} + (1 - \rho_i) \left(\phi_\pi \pi_t^{CPI} + \phi_X \hat{X}_t \right). \quad (21)$$

Here hats denote log deviations from the stationary equilibrium, so this is the compact first-order representation of the inertial policy rule.

Wages and profits feed household resources, while monetary policy affects the return on saving. The model gathers these financial flows in a single intermediary before closing the external and fiscal accounts.

Mutual fund. Households hold one liquid claim issued by a competitive mutual fund. Behind that claim, the fund owns domestic equity J_t , government debt B_t , and net foreign assets NFA_t , so that $A_t = J_t + B_t + NFA_t$. This arrangement keeps the household problem one-dimensional while allowing aggregate saving to finance firms, the government, or the rest of the world.

Domestic equity equals the present value of future dividends. Together with bond holdings and valuation changes, these dividends determine the realized return on the liquid claim that appeared in the household budget constraint.

External balance. The foreign component of the fund evolves with the trade balance. A real uncovered-interest-parity condition links the domestic return to the foreign return and the expected evolution of q_t ; the baseline does not rely on a debt-elastic risk premium. Consumption imports and imported inputs enter the external account at the border price q_t , because tariff payments remain inside the domestic economy. This distinction separates the country's resource transfer to foreign suppliers from the government's tariff revenue.

That revenue closes the remaining institutional block. Its treatment matters for incidence because an immediate rebate would partly undo the household purchasing-power loss on impact.

Fiscal policy. Tariff revenue is

$$\mathcal{R}_t^T = \tau_t^C q_t C_{M,t} + \tau_t^I q_t M_{I,t}. \quad (22)$$

In the baseline, tariff revenue initially reduces public debt, after which a gradual labor-income tax rule brings debt back to its stationary level. Thus fiscal adjustment stabilizes debt without imposing an immediate transfer that mechanically offsets the purchasing-power loss. A uniform household-rebate closure is considered separately in the robustness analysis.

Taken together, these blocks describe the household, production, policy, and external sides of the economy. They can now be assembled into a compact definition of equilibrium.

Definition 1 (Equilibrium). Given the initial household distribution $\mathcal{D}_0$, initial asset positions, and the exogenous foreign, productivity, and tariff processes, an equilibrium is a sequence of household policies and distributions, prices and wages, production and trade quantities, policy variables, and financial positions with the following properties. Households solve equations (13)–(14), allocate expenditure as summarized by equations (15)–(17), and generate a distribution consistent with their saving policies. Firms optimize subject to the production technology in equation (19), while wage setting and monetary policy satisfy equations (20)–(21). The mutual fund, fiscal authority, and external sector satisfy their respective budget constraints, with tariff revenue given by equation (22) and uncovered interest parity holding. Finally, all domestic-good, labor, asset, fiscal, and external markets clear.

The definition closes the model, but it does not yet reveal how the incidence mechanism enters. I make that connection in two steps. First, I recover the household-exposure statistic while holding the remaining equilibrium margins fixed. I then locate that object within the full general-equilibrium system.

Household incidence in partial equilibrium. By Shephard’s lemma, the local expenditure required to compensate household i for a change in the consumption-import tariff satisfies

$$\alpha_{i,t}^\tau \equiv \frac{\partial \log x_{i,t}}{\partial \vartheta_t^C} = \kappa_{\tau,t} s_{M,i,t}, \quad \kappa_{\tau,t} \equiv \frac{\partial \log p_{M,t}^C}{\partial \vartheta_t^C}. \quad (23)$$

The household import share therefore becomes the model counterpart of the exposure measure introduced in Section 2. Neither side of the incidence relationship is imposed directly. Exposure varies with expenditure, permanent basket type, and equilibrium relative prices, while spending propensities emerge from assets, productivity, patience, and the borrowing constraint.

To capture the timing of the shock, let $M_{z,0s}$ denote the date-0 expenditure response of a household in stationary state z to one unit of liquid resources received at date s . Likewise, let q_s^τ denote the exogenous tariff-wedge path per unit innovation, normalized so that $q_0^\tau = 1$ and excluding the endogenous border-price response. For the AR(1) process used below, $q_s^\tau = \rho_\tau^s$. These two objects combine in the tariff-relevant iMPC,

$$\Psi_z(q^\tau) = \sum_{s \geq 0} M_{z,0s} q_s^\tau. \quad (24)$$

Holding initial baskets, household distributions, labor income, returns, transfers, and all non-import prices fixed, the fixed-basket, resource-equivalent purchasing-power component satisfies

$$\frac{d\mathcal{S}_0^{PP}}{\mathcal{S}} = -\mathbb{E}^S [\Psi_z(q^\tau) \alpha_z^\tau] d\vartheta_0^C. \quad (25)$$

Appendix B.4 provides the proof. One measurement detail is important here. The abstract statistic is defined over real consumption expenditure, which in the Stone–Geary block is $x_z = \zeta + \mathcal{P}_\xi c_z$, rather than the composite quantity c_z . Equation (25) is thus the expenditure-based counterpart of the earlier result: its base, weights, and resource response all refer to x_z . At fixed prices, $dx_z = \mathcal{P}_\xi dc_z$, and the stationary normalization sets $\mathcal{P}_\xi = 1$; nevertheless, expenditure continues to include the subsistence cost. For this reason, the equation should not be read as an identity for the composite-consumption index.

In the quantitative analysis, I evaluate the full path-weighted object under the same AR(1) tariff process used in general equilibrium. For comparison, I also report the contemporaneous special case, $q_0^\tau = 1$ and $q_s^\tau = 0$ for $s > 0$, for which $\Psi_z = M_{z,00}$. The first calculation incorporates anticipated future purchasing-power losses, whereas the second isolates impact incidence. In both cases, the HANK model supplies the state-level dated transfer Jacobians and the joint distribution of exposure and intertemporal spending responses.

For an observable partition $b(z)$, the same statistic inherits the average, vertical, and horizontal decomposition in equation (12). The implementation groups households by population expenditure decile, so V_τ captures covariance across decile means and H_τ captures the residual covariance within deciles. Neither component is itself an equilibrium comparison between two model economies.

The distinction is substantive: exposure dispersion produces unequal household burdens, but it strengthens aggregate expenditure only when the more exposed households also have stronger spending responses.

From household incidence to general equilibrium. The partial-equilibrium equation isolates the mechanism, but the tariff also moves the prices and incomes held fixed in that calculation. To see how the same household block enters the equilibrium response, stack the endogenous aggregate sequences in $\mathbf{y}$, household aggregates in $\mathbf{h}$, and exogenous shocks in $\mathbf{z}$. Let $\mathbf{J}_{\mathbf{y}}^H$ and $\mathbf{J}_{\mathbf{z}}^H$ denote the corresponding total derivatives of the household block, and let $\mathcal{F}(\mathbf{y}, \mathbf{h}, \mathbf{z}) = 0$ collect the remaining equilibrium conditions. Provided that the local equilibrium Jacobian is nonsingular, the linearized response is

$$d\mathbf{y} = - [\mathcal{F}_{\mathbf{y}} + \mathcal{F}_{\mathbf{h}}\mathbf{J}_{\mathbf{y}}^H]^{-1} [\mathcal{F}_{\mathbf{z}} + \mathcal{F}_{\mathbf{h}}\mathbf{J}_{\mathbf{z}}^H] d\mathbf{z}. \quad (26)$$

The sufficient statistic gives the mathematical intuition for the household-exposure margin contained in this system. With baskets, incomes, returns, transfers, and non-import prices fixed, purchasing-power losses are more contractionary when exposure is concentrated among households with stronger intertemporal spending responses. Yet the statistic is not a separate entry in equation (26). Once equilibrium prices and incomes move, the tariff generates a vector of reoptimized household responses that interacts with wages, employment, returns, fiscal and monetary policy, external adjustment, and the imported-input channel. The equation is therefore a local implicit-function representation of equilibrium—not an additional counterfactual, a mapping from the sufficient statistic to output, or an identified output decomposition. Appendix B.17 reports the underlying chain rule and equilibrium system.

The quantitative exercise therefore asks how alternative structures of household exposure change this equilibrium, rather than treating the partial-equilibrium statistic as an output response. Three model economies provide the comparison.

Three exposure economies. The comparison begins with a *common-basket economy*. It preserves the total steady-state subsistence cost but divides its domestic and imported components, as well as the supernumerary weight, in the aggregate import proportion. Every household consequently has the same stationary import share. Incomplete markets, the intertemporal household problem, and MPC heterogeneity remain in place; only exposure heterogeneity is removed.

The *vertical-exposure economy* then activates the Stone–Geary subsistence quantities while keeping a common supernumerary import weight. Exposure now varies along the expenditure gradient, but not with permanent basket type conditional on expenditure. Finally, the *full-exposure economy* combines that gradient with permanent variation in ω_{ξ} and thereby adds residual basket-type dispersion.

Across the three economies, I hold fixed the aggregate final-import share and every parameter unrelated to exposure, while recomputing each household block and its sequence-space Jacobians. At the normalized stationary relative import price, the systems share the same supernumerary price index and total subsistence cost. Their intertemporal budget sets therefore coincide, as do

their composite-consumption and saving policies and their marginal stationary distributions over intertemporal states. What differs away from the stationary equilibrium are the exposure-specific Jacobians; the full economy also retains the independent basket-type label.

Thus these are equilibrium counterfactuals, not *ex post* reweighting exercises. Moving from common to vertical baskets isolates the expenditure gradient; moving from vertical to full baskets measures the incremental equilibrium effect of permanent basket-type dispersion. That second difference should not be confused with H_τ , the within-group covariance term in the sufficient-statistic identity.

Solution and quantitative experiments. I solve the stationary household problem with the endogenous-grid method (EGM) and compute aggregate transition dynamics using sequence-space Jacobians following [Auclert et al. \(2021a\)](#). Each of the common-basket, vertical-exposure, and full-exposure economies receives its own stationary equilibrium and household Jacobians.

In the benchmark, households can immediately reallocate expenditure between domestic and imported final goods through the Stone–Geary CES demand system. The robustness exercises then vary the substitution elasticity, tariff persistence, imported-input intensity, nominal rigidity, fiscal closure, and the timing of intratemporal basket adjustment. Finally, because aggregate transitions are solved to first order, the reported responses scale linearly with the innovation by construction. Numerical residuals therefore diagnose the linear system, rather than the approximation error of a nonlinear transition.

4 Quantitative design

The quantitative exercise is a quarterly, stylized benchmark for an advanced small open economy rather than an estimate for a particular country. Stationary domestic value added is normalized to one, and the aggregate architecture and principal assignments follow the open-economy HANK literature, especially [Auclert et al. \(2021b\)](#) and [Druedahl et al. \(2024\)](#). Throughout this section, I distinguish documented targets, direct parameter assignments, external empirical benchmarks, and implications generated by the model. The numerical magnitudes illustrate the mechanism and its sensitivity; they should not be read as estimates for an average OECD economy.

The benchmark parameterization determines the marginal distributions of import exposure and household spending responses. It does not estimate their complete joint distribution from a linked household sample; the dependence between them is therefore reported transparently as a model implication and subjected to sensitivity analysis.

The three economies share the same aggregate calibration and stationary final-import share. Their household blocks and sequence-space Jacobians are computed separately. The stationary household policies and distributions coincide numerically in the frictionless steady state, whereas the response Jacobians differ across exposure systems. Differences across them are equilibrium counterfactuals, not *ex post* reweighting exercises.

4.1 Calibration

Calibration strategy. The household block is calibrated in two steps. First, three permanent discount-factor types and their population masses are chosen jointly to match household financial wealth and the average quarterly MPC. Second, the Stone–Geary parameters reproduce the aggregate final-import share and generate a moderate expenditure gradient and residual basket dispersion in the baseline. The resulting standard deviation of expenditure-decile mean import shares is 0.0315, and the mean within-decile standard deviation is 0.0250. These two values are baseline parameterization choices, not independently documented empirical targets and not moments attributed to a particular country. Section 4.2 confronts them with a separate Spanish-data validation and reports a Spanish-EPF sensitivity. The remaining parameters are literature-based assignments or direct normalizations.

Permanent basket weights satisfy

$$\omega_{\xi} = \frac{\exp(\omega_0 + \sigma_{\xi}\xi)}{1 + \exp(\omega_0 + \sigma_{\xi}\xi)}, \quad (27)$$

where ξ takes five standardized values. The baseline sorting parameter between patience and basket type is zero. The model therefore does not impose that intrinsically impatient households have more import-intensive tastes. Table 1 reports the structural calibration.

Numerical implementation. The idiosyncratic productivity process is approximated by a seven-state Rouwenhorst chain and the household distribution contains three permanent discount-factor types and five permanent basket types.

The common-basket, vertical-exposure, and full-exposure economies are solved independently. The common economy sets both its subsistence composition and its supernumerary import weight to the aggregate final-import share. In the vertical counterfactual, the common supernumerary import weight is adjusted so that all three economies reproduce the same aggregate final-import expenditure share.

Moments. Table 2 compares the stationary model with its aggregate targets and assignments. Panel A contains the objects used in calibration, while Panel B reports three untargeted model implications. The benchmark exposure dispersion is deliberately excluded: it follows from the basket parameterization and is not an independently observed target.

The model matches the targeted moments closely. The small difference in government consumption arises because the steady-state government budget, dividends, financial wealth, and the fixed operating cost are solved jointly.

The model produces a four-quarter cumulative MPC of 0.284. This object sums the first four responses after an impact transfer, whereas the tariff-relevant iMPC uses the date-0 response to transfers dated along the tariff path. These are, respectively, column and row slices of the expenditure–transfer Jacobian and need not coincide. The cumulative response, the disposable-flow-income Gini, and the low-liquidity share are reported as untargeted implications rather than compared

Table 1: Calibration

Parameter	Description	Value	Source / target
<i>Panel A: Preferences and household heterogeneity</i>			
σ	Relative risk aversion	1.000	Standard
φ	Inverse Frisch elasticity	2.000	Standard
$\{\beta_L, \beta_M, \beta_H\}$	Discount-factor types	$\{0.96891, 0.98800, 0.99375\}$	Wealth and average MPC
$\{\pi_L^\beta, \pi_M^\beta, \pi_H^\beta\}$	Population masses	$\{0.20, 0.35, 0.45\}$	Wealth and average MPC
ρ_e	Earnings persistence	0.966	Standard
σ_e	Earnings dispersion	0.500	Standard
<i>Panel B: Household consumption baskets</i>			
η_C	Elasticity between domestic and imported goods	1.500	Standard
$\bar{c}_H$	Domestic subsistence requirement	0.02336	Benchmark exposure gradient
$\bar{c}_M$	Imported subsistence requirement	0.05664	Benchmark exposure gradient
ω_0	Mean import taste	-1.15422	Aggregate import share
σ_ξ	Basket-type dispersion	0.14062	Benchmark basket dispersion
$\chi_{\beta, \xi}$	Patience-basket sorting	0.000	Baseline
<i>Panel C: Production and international trade</i>			
μ_p	Gross product markup	1.200	Standard
ν_I	Elasticity between value added and imported inputs	0.500	Auclert et al. (2021b)
η_X	Export-demand elasticity	2.000	Standard
<i>Panel D: Nominal, monetary, fiscal, and external blocks</i>			
κ_W	Wage Phillips-curve slope	0.050	Druehl et al. (2024)
ρ_i	Interest-rate smoothing	0.900	Standard
ϕ_π	Response to CPI inflation	1.500	Standard
ϕ_X	Response to domestic value added	0.000	Baseline
ψ_B	Labor-tax response to debt	0.020	Debt stabilization
$\bar{\tau}^\ell$	Steady-state labor-income tax rate	0.2700	Stationary government budget
ϕ_{NFA}	External risk-premium elasticity	0.000	Perfect capital mobility
<i>Panel E: Tariff process</i>			
ρ_τ	Tariff persistence	0.750	Baseline
$\varepsilon_{\tau, 0}$	Impact log tariff-wedge innovation	0.100	First-order graphical normalization

Notes: The model is quarterly. Parameters calibrated to moments refer to the targets reported in Table 2. The innovation of 0.100 is a first-order graphical normalization of the log tariff-wedge shock. From a zero initial tariff, it translates mechanically into an exact ad valorem rate of $\exp(0.10) - 1 = 10.52\%$, but the plotted responses are not a separate nonlinear simulation of that reform. The fixed operating cost is determined residually and equals 0.167 of stationary GDP.

with undocumented reference values. The path-weighted sufficient statistic uses the model’s dated transfer responses directly rather than inferring persistence from a four-quarter comparator. None of the central tariff results relies on targeting an unusually large cumulative MPC.

Table 2: Aggregate calibration and model counterparts

Moment	Benchmark value	Model
<i>Panel A: Calibration targets and assignments</i>		
Annual real interest rate	0.0200	0.0200
Average quarterly MPC	0.2000	0.2000
Household financial wealth / quarterly GDP	10.0000	10.0000
Government consumption / GDP	0.2000	0.2029
Government debt / quarterly GDP	2.3200	2.3200
Total imports / GDP	0.4200	0.4200
Imported inputs / total imports	0.5500	0.5500
Gross product markup	1.2000	1.2000
Final-import expenditure share	0.3000	0.3000
<i>Panel B: Untargeted model implications</i>		
Four-quarter cumulative MPC	—	0.2836
Disposable-flow income Gini	—	0.2751
Share with very low liquid assets	—	0.2158

Notes: Panel A reports calibration targets and direct assignments. Dashes in Panel B indicate model implications that are reported but not targeted. Very low liquid wealth is defined as assets below 5% of mean household flow income. Ratios are expressed as fractions rather than percentages.

4.2 External validation and empirical scope

As an external validation, I combine the 2022 Spanish Household Budget Survey (EPF) with INE input–output tables to reconstruct direct-final import exposure. This exercise evaluates whether the benchmark exposure structure spans empirically plausible distributions; it does not select Spain as the model economy or replace the benchmark calibration. Between-expenditure-decile dispersion is similar in the reconstruction and the benchmark (0.0331 and 0.0315), whereas within-decile dispersion is much larger (0.0959 and 0.0250). I therefore use the Spanish evidence only to construct a high-dispersion sensitivity. Its central node sets the basket-dispersion coefficient to 0.644, 4.58 times the benchmark value of 0.14062, while re-centering the taste intercept to preserve mean import exposure. Appendix C documents the weighting, mapping, uncertainty, and alternative definitions.

The available Spanish sources do not identify the joint distribution required by the sufficient statistic. EPF disciplines the marginal distribution of imputed import exposure but contains no household MPC or iMPC. The Survey of Household Finances (EFF) contains balance-sheet variables and a twelve-month stated spending-response proxy, but not the product-level expenditure basket required to observe import exposure. Because the samples cannot be linked, the EPF–EFF exercise identifies only the covariance between conditional means based on common observables, which is small and negative (−0.000608 with expenditure weights). It does not identify the resid-

ual exposure–response covariance. Rearrangement bounds that preserve the conditional marginal distributions span impact amplification from -43.6% to 46.3% . The model’s 11.3% impact-only result lies inside this set but is not selected by the available data; the 7.0% path-weighted result is likewise a conditional model implication, not an empirical estimate.

4.3 Steady-state model implications

Figure 2 shows the joint cross-sectional structure that drives the tariff mechanism. The left panel shows a strong expenditure gradient. Final imports account for 38.4% of expenditure in the bottom decile and 27.5% in the top decile. The common-basket economy removes this gradient and sets every household’s exposure to the aggregate share of 30% . The vertical and full economies have almost identical decile means because permanent basket types are parameterized to generate dispersion conditional on expenditure, not to change the average gradient across expenditure deciles.

The middle panel shows equally strong heterogeneity in spending responses. The quarterly iMPC is close to one in the bottom liquid-asset decile, falls below 0.12 in the third decile, and approaches zero at the top of the distribution. The average quarterly MPC of 0.20 masks a highly concentrated distribution of short-run responses.

The two forms of heterogeneity are positively aligned in the stationary model. As shown in the right panel, average final-import exposure rises from approximately 28% in the lowest-iMPC decile to 37% in the highest. The expenditure-weighted correlation between exposure and the quarterly iMPC is 0.439 . This is a model-generated correlation, not a correlation estimated from a linked household dataset.

This sorting is not imposed through basket tastes. It emerges because low-expenditure households tend to hold fewer liquid assets and devote a larger share of expenditure to necessity-like imports. The model therefore generates the positive covariance required by the sufficient statistic before any aggregate tariff experiment is run.

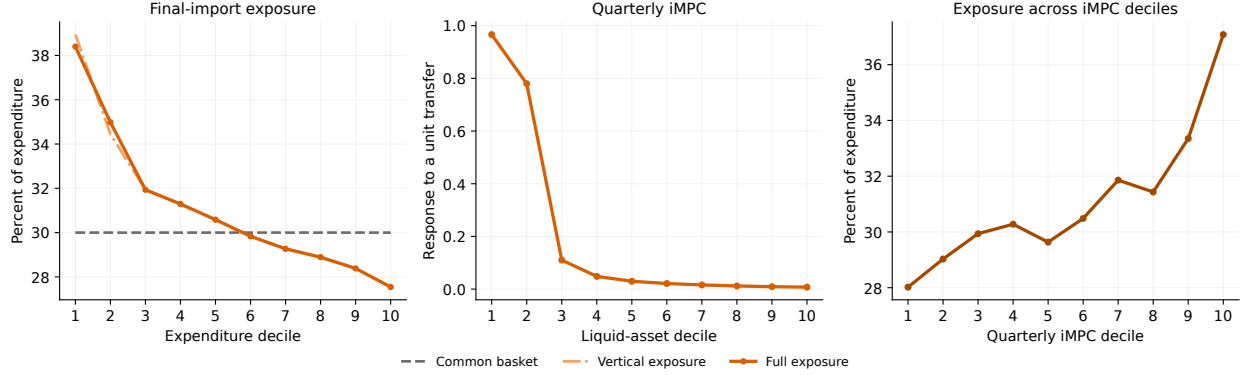


Figure 2: Steady-state household heterogeneity. The left panel reports model final-import expenditure shares across expenditure deciles; the aggregate share is targeted, whereas the dispersion follows from the benchmark parameter assignment. The middle panel is a model-implied profile of impact iMPCs across liquid-asset deciles; only the population-average quarterly MPC is targeted. An impact iMPC is the date-0 expenditure response to a one-time unit liquid transfer at date 0, holding productivity income, the asset return, and the import price at stationary values. The right panel is the untargeted model implication for average final-import exposure across impact-iMPC deciles in the full-exposure economy.

5 Results

The quantitative analysis proceeds from aggregate transmission to household incidence and, finally, to the role of sorting in aggregate demand. This order separates three objects that need not move together: the size of the aggregate contraction, the distribution of household losses, and the covariance between those losses and households’ spending responses. Unless otherwise stated, the figures scale local first-order responses by a 0.10 log-wedge innovation.

Aggregate transmission. Figure 3 shows the familiar macroeconomic side of the shock. Domestic value added and consumption fall, while inflation rises initially. The responses are front-loaded: inflation reverses as the tariff wedge fades, whereas quantities and the real wage recover more gradually. This contractionary and inflationary pattern is standard when tariffs raise production costs and induce expenditure switching (Barattieri et al., 2021; Auclert et al., 2025).

The two tariff margins nevertheless perform different tasks. The tariff on imported inputs accounts for most of the decline in domestic activity because foreign inputs are difficult to replace in the short run. On the other hand, the tariff on final goods mainly reduces import demand and creates the household cost-of-living shock studied below. To first order, the two experiments add up.

This distinction matters for interpretation: The intermediate-input margin largely determines the aggregate production response, whereas the final-goods margin supplies the direct household exposure that generates distributional effects. The mechanisms interact in general equilibrium through wages, employment, returns, and monetary policy, but the figure shows that their first-order contributions can still be read separately.

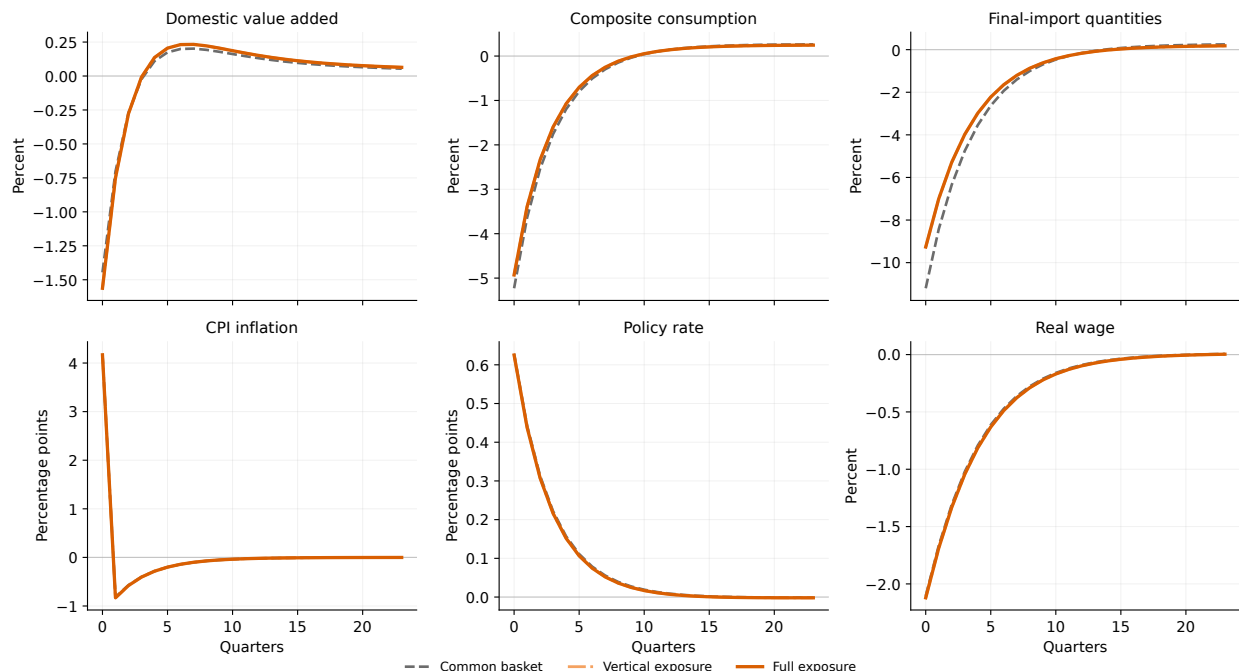


Figure 3: Aggregate responses to a joint tariff shock. The figure reports local first-order impulse responses scaled to a 0.10 innovation in the log tariff wedge applied jointly to imported final consumption goods and imported intermediate inputs. This is a graphical normalization of linear semi-elasticities, not a nonlinear policy simulation. Real quantities are percentage deviations from stationary values. CPI inflation and the policy rate are expressed in percentage points.

Distributional incidence and household responses. The aggregate response does not reveal the distributional side of the shock. Figure 4 therefore follows all ten stationary expenditure deciles from impact through quarter 12. The cost-of-living burden is ordered clearly: low-expenditure households devote a larger share of their initial baskets to imports, so the same price movement absorbs more of their resources. This ordering fades as the tariff wedge dissipates, but its initial gradient is persistent enough to create a meaningful difference in household purchasing power.

Interestingly, consumption does not inherit the same ordering, as the right-side panel of Figure 4 shows non-monotonic responses. Expenditure ranks households by their baskets more cleanly than it ranks their capacity to smooth a loss: liquid wealth, borrowing constraints, and discount-factor types also vary across deciles. The broad contrast between the lower tail and the rest of the distribution is pronounced in the benchmark, while the smaller reversals between adjacent deciles reflect the composition of household states within each group.

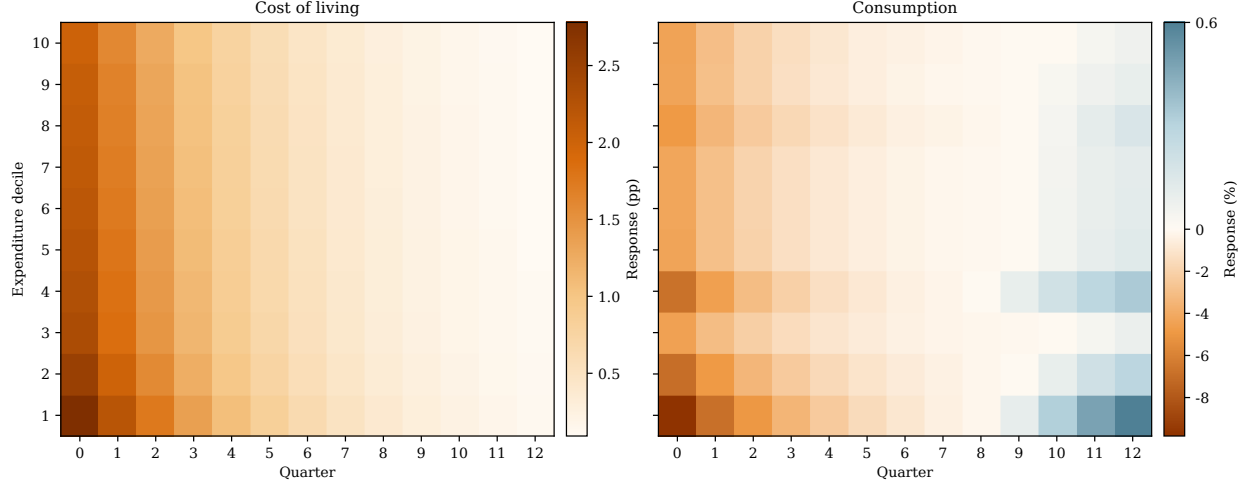


Figure 4: Household incidence across expenditure deciles. Each cell reports the response in the full-exposure economy of a group fixed by stationary expenditure under the 0.10 first-order normalization. The left panel shows the log cost-of-living response in percentage points; the right panel shows the percentage deviation of group composite consumption from its stationary mean. Group membership and normalized population weights are held fixed over the transition. These responses are not welfare measures.

The two panels also make clear why cost-of-living incidence and consumption cannot be treated as interchangeable measures. The first is a direct price object pinned down by the initial basket. The second is an equilibrium response that combines this price pressure with households’ financial positions and with changes in labor and asset income. Permanent basket types add little to the average response but widen the cross-sectional distribution of cost-of-living changes. Accordingly, the vertical–full comparison is most informative about incidence. The detailed comparison, including the Spanish-data sensitivity, remains in Appendix C.

Figure 5 makes the consumption accounting explicit. For expenditure decile d , let $\mathcal{H}_{d,0}^{\text{CB}}[\cdot]$ denote the linear map from a complete anticipated household-input path in the common-basket economy to impact composite consumption, normalized by that decile’s stationary consumption. The set $\mathcal{K}$ contains the import-price, wage, hours, labor-tax, and financial-return components reported in the figure. Linearity gives

$$\hat{c}_{d,0}^{\text{CB}} = \sum_{k \in \mathcal{K}} \mathcal{H}_{d,0}^{\text{CB}} \left[d\mathbf{u}_{\text{CB}}^k \right], \quad \mathcal{K} = \{p_M, \text{wage}, \text{hours}, \text{tax}, r^a\}. \quad (28)$$

Here each $d\mathbf{u}_{\text{CB}}^k$ is the entire solved transition path associated with that input component, even though the left-hand side records the response at impact. Thus, the bars incorporate anticipation and household dynamics rather than only contemporaneous movements in prices or income.

The full-exposure response is closed by a residual defined as the difference between the full and

common-basket economies:

$$\hat{c}_{d,0}^F = \underbrace{\sum_{k \in \mathcal{K}} \mathcal{H}_{d,0}^{\text{CB}} \left[du_{\text{CB}}^k \right]}_{\text{standard household-input components}} + \underbrace{(\hat{c}_{d,0}^F - \hat{c}_{d,0}^{\text{CB}})}_{\mathcal{R}_{d,0}^{\text{exp}}}. \quad (29)$$

The black point in the figure is $\hat{c}_{d,0}^F$ (the total effect), the first bracket contains the five standard bars, and $\mathcal{R}_{d,0}^{\text{exp}}$ is the exposure bar. This last term is a transparent full-minus-common residual. It includes both the change in the household exposure map and any induced change in equilibrium input paths, so it should not be interpreted as an orthogonal causal shock or as the contribution of permanent basket types alone.

The full-minus-common exposure residual varies across deciles rather than shifting every group in the same direction. It deepens the contraction in the first two deciles, changes sign at the third, is close to zero in the fourth, and generally cushions the response higher in the expenditure distribution. This sign change is central to the distributional result: heterogeneous exposure does not simply scale a common loss but reallocates it across households. Import prices and labor income nevertheless remain large contributors at the bottom. Their interaction with direct price exposure is characteristic of heterogeneous-agent transmission (Kaplan et al., 2018; Auclert et al., 2021b).

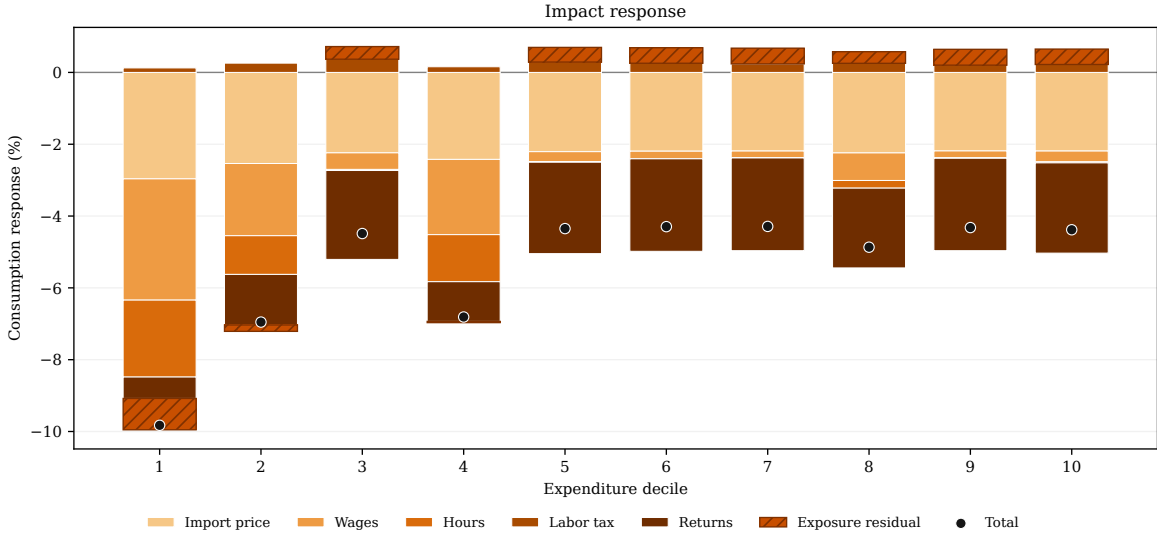


Figure 5: Impact consumption response by expenditure decile and accounting component. Values are percentage-point contributions to each decile’s composite-consumption response under the 0.10 normalization. Negative bars deepen the contraction and positive bars cushion it; the black point is their algebraic sum. The five standard components apply, one at a time, the complete anticipated input paths from the solved common-basket equilibrium through its household block. *Import price* combines cost-of-living, intratemporal-composition, and intertemporal responses; *Returns* denotes the realized mutual-fund return. *Exposure residual* is the full-minus-common residual, not an identified causal channel. Group membership is fixed by stationary expenditure, and benchmark transfers are zero.

Sorting and general-equilibrium amplification. The channel accounting describes how the solved equilibrium reaches different households. Table 3 isolates a narrower object: the extent to which sorting amplifies the direct purchasing-power effect when baskets and non-price household inputs are held fixed, before general-equilibrium feedback. It therefore complements rather than duplicates Figure 5.

Table 3: Path-weighted sufficient statistic by tariff persistence

Tariff persistence	Average	Between	Within	Total	Amplification (%)
0.00 (impact only)	−0.382	−0.041	−0.003	−0.425	11.28
0.50	−0.472	−0.041	−0.002	−0.516	9.22
0.75 (central)	−0.624	−0.042	−0.002	−0.668	7.01
0.90	−0.966	−0.041	−0.001	−1.008	4.40

Notes: The first four numerical columns are percentage-point effects of a 0.10 log tariff-wedge innovation. Expenditure deciles are defined once on the common (β, e, a) grid using fractional population memberships and replicated identically across permanent basket types ξ ; expenditure weights are then used inside the statistic. The impact-only row sets $\rho_\tau = 0$; the central process has $\rho_\tau = 0.75$. Amplification is the percentage increase relative to the product of mean exposure and mean intertemporal spending response. These are conditional model implications under the joint distribution generated by the baseline parameterization. Source: Model calculations.

Under the central tariff path, sorting makes the direct loss 7.0% larger in magnitude than the product-of-marginals benchmark. Almost all of that increment comes from sorting across expenditure groups; the within-group covariance is small. This result is parsimonious: the key departure from a representative exposure calculation is the systematic alignment between import shares and intertemporal spending responses across the expenditure distribution. The percentage amplification declines with tariff persistence because the average path effect grows relative to the covariance component.

The statistical decomposition and the model-economy comparisons remain distinct. Permanent basket types can widen household incidence without creating an additional covariance term because basket type ξ is independent of the intertemporal spending response in the benchmark. Conversely, the vertical–full economy comparison measures the incremental equilibrium effect of those types, not the within-group term in the table.

Nor does the direct statistic map one for one into aggregate outcomes. Impact output falls more under vertical and full exposure than under a common basket, but the two heterogeneous-basket economies are almost indistinguishable in the aggregate. The expenditure gradient therefore accounts for nearly all of the baseline aggregate exposure effect, whereas permanent basket types mainly redistribute the household burden. Inflation, the policy rate, and the real wage barely differ across the three economies, which helps locate the result in household expenditure composition rather than in a different aggregate policy response.

Composite consumption and domestic value added need not move in the same direction across economies. Highly exposed households substitute away from necessity-like imports less aggressively.

They preserve more imported consumption but redirect less expenditure toward domestic goods, so composite consumption falls less even as domestic activity falls more. The full–common output gap is conditional on frictionless basket adjustment and becomes negligible as composition approaches the fixed-basket limit in Appendix D.3.

The distinction is therefore sharp. Exposure heterogeneity matters strongly for the distribution of household losses, but more moderately for the size of the aggregate recession. The table quantifies the direct sorting force, the bars show how it interacts with standard household-input components, and the economy comparisons reveal how much survives general equilibrium. None of these responses is a welfare ranking: wages, employment, returns, taxes, and fiscal policy also adjust.

6 Conclusion

This paper shows why average household import exposure is not sufficient for understanding the demand effects of a tariff. Even before production and policy adjust, the direct purchasing-power effect depends on whether the households facing the largest cost-of-living losses are also those with the strongest spending responses.

The path-weighted sufficient statistic makes this argument precise by separating average exposure from covariance across and within observable expenditure groups. Under the benchmark persistent tariff process, sorting raises the direct loss by 7.0% relative to the product of mean exposure and mean intertemporal spending response. Most of this amplification occurs across expenditure groups, linking the aggregate-demand effect to the systematic alignment of import shares, liquid wealth, and spending responses.

The quantitative model then shows how general equilibrium reshapes this direct force. Tariffs on imported inputs generate most of the contraction in domestic activity, whereas those on final consumption goods create the household purchasing-power shock. Across expenditure deciles, consumption responses are not monotone because direct price exposure interacts with wages, hours, fiscal adjustment, and asset returns. Heterogeneous exposure deepens the impact loss at the bottom of the expenditure distribution while cushioning much of the upper part. It therefore reallocates the household burden rather than simply scaling a common consumption response.

The same distinction appears when the three exposure economies are compared. The expenditure gradient accounts for nearly all of the baseline aggregate exposure effect, while permanent basket-type dispersion mainly widens the distribution of cost-of-living changes among otherwise similar households. Moreover, the additional aggregate output effect becomes negligible as basket composition approaches the fixed-basket limit. Exposure heterogeneity is thus more consequential for incidence than for the size of the aggregate recession.

The quantitative benchmark is illustrative rather than an estimate for a particular country. Its natural empirical use is as a portable framework whose exposure and spending-response distributions can be replaced when suitable linked country data become available.

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A Partial-equilibrium regularity conditions and proofs

This appendix states the regularity conditions underlying the partial-equilibrium incidence results and provides formal proofs of Propositions 1 and 2 and Corollaries 1 and 2.

A.1 Measure, price path, integrability, and units

Let $(\mathcal{I}, \mathcal{F}, \mu)$ be the household measure space. Initial real expenditure c_i is strictly positive and measurable, with

$$0 < C = \int_{\mathcal{I}} c_i d\mu(i) < \infty.$$

The expenditure-weighted probability measure is

$$\mathbb{P}^C(A) = \frac{1}{C} \int_A c_i d\mu(i), \quad A \in \mathcal{F}. \quad (30)$$

Expectations and covariances under this measure are denoted by E^C and Cov^C .

The consumer-price derivative is assumed to factorize as

$$d \log p_{g,s} = \theta_g q_s^\tau d\tau_0, \quad q_0^\tau = 1, \quad (31)$$

where the deterministic normalized path $q^\tau = \{q_s^\tau\}_{s \geq 0}$ is common across goods and cross-good heterogeneity enters through θ_g . This separability is what permits tariff exposure to be summarized by the scalar $\alpha_i^\tau = \sum_g \omega_{ig} \theta_g$. With good-specific normalized paths, the same aggregation logic would apply to a path-dependent household exposure rather than to this scalar.

For each household, the reference-basket expenditure index is differentiable at the initial price vector and

$$d \log P_{i,s} = \sum_{g \in \mathcal{G}} \omega_{ig} d \log p_{g,s}.$$

If $\mathcal{G}$ is countable, I additionally assume $\sum_{g \in \mathcal{G}} \omega_{ig} |\theta_g| < \infty$ almost everywhere. These conditions are automatic when $\mathcal{G}$ is finite and the good-level price derivatives are finite.

Tariff exposure α_i^τ is measurable and essentially bounded under $\mathbb{P}^C$: there is a finite constant A_τ such that

$$|\alpha_i^\tau| \leq A_\tau \quad \text{for } \mathbb{P}^C\text{-almost every } i. \quad (32)$$

For every $s \geq 0$, $M_{i,0s}$ is measurable, and

$$E^C \left[\sum_{s=0}^{\infty} |M_{i,0s} q_s^\tau| \right] < \infty. \quad (33)$$

For $\mathbb{P}^C$ -almost every household, date-0 expenditure is directionally differentiable at the steady state

along the dated liquid-resource direction $h_{i,s} = -\alpha_i^\tau c_i q_s^\tau$, with derivative

$$Dc_{i,0}[h_i] = \sum_{s=0}^{\infty} M_{i,0s} h_{i,s}.$$

The associated household difference quotients are dominated, after division by c_i , by an integrable function in a neighborhood of the steady state. This condition permits differentiation under the household integral. Together with the absolute-summability condition above, it also permits interchanging the household integral and the time sum. These conditions imply that

$$\Psi_i(q^\tau) = \sum_{s=0}^{\infty} M_{i,0s} q_s^\tau$$

exists almost everywhere and belongs to $L^1(\mathbb{P}^C)$. They also imply

$$E^C \left[|\alpha_i^\tau| \sum_{s=0}^{\infty} |M_{i,0s} q_s^\tau| \right] \leq A_\tau E^C \left[\sum_{s=0}^{\infty} |M_{i,0s} q_s^\tau| \right] < \infty. \quad (34)$$

Hence Fubini's theorem applies to the household integral and the time sum, and both $\Psi_i(q^\tau)\alpha_i^\tau$ and all covariances used below are integrable.

The units are consistent as follows. Both $c_{i,0}$ and $y_{i,s}$ are measured in the same real-expenditure units, so $M_{i,0s}$ is unit-free. The normalized path q_s^τ , the price semi-elasticity θ_g , the expenditure shares ω_{ig} , exposure α_i^τ , and the log-wedge innovation $d\tau_0$ are dimensionless. Therefore $\Psi_i(q^\tau)$ and $s_\tau(q^\tau)$ are dimensionless, as is the normalized response dC_0^{PP}/C .

The identities below do not require sign restrictions. For their usual economic interpretation as a fall in spending after a tariff increase, sufficient conditions are

$$d\tau_0 > 0, \quad \theta_g \geq 0, \quad q_s^\tau \geq 0, \quad M_{i,0s} \geq 0$$

almost everywhere. These conditions imply $\alpha_i^\tau \geq 0$, $\Psi_i(q^\tau) \geq 0$, and $s_\tau(q^\tau) \geq 0$. If any sign restriction is relaxed, the same equations remain valid but describe a signed spending response.

For the group decomposition, let $b : \mathcal{I} \rightarrow \mathcal{B}$ be a measurable finite or countable partition and define

$$\pi_b^C = \mathbb{P}^C \{i : b(i) = b\}. \quad (35)$$

Groups with $\pi_b^C = 0$ are omitted. Conditional moments are taken under $\mathbb{P}^C$, and

$$\text{Cov}_b^C(X_i, Y_i) = E_b^C [(X_i - E_b^C[X_i])(Y_i - E_b^C[Y_i])].$$

The preceding conditions ensure that these conditional covariances and their average under $\mathbb{P}^C$ are well defined.

A.2 Proof of the demand sufficient statistic

Proof of Proposition 1. The Divisia derivative of the cost of purchasing household i 's reference basket is

$$\begin{aligned} d \log P_{i,s} &= \sum_{g \in \mathcal{G}} \omega_{ig} d \log p_{g,s} \\ &= \left(\sum_{g \in \mathcal{G}} \omega_{ig} \theta_g \right) q_s^\tau d\tau_0 \\ &= \alpha_i^\tau q_s^\tau d\tau_0. \end{aligned}$$

Under the normalization $P_{i,ss} = 1$, the corresponding first-order increase in expenditure on the reference basket is $\alpha_i^\tau c_i q_s^\tau d\tau_0$. The equivalent liquid-resource perturbation is therefore

$$dy_{i,s}^{eq} = -\alpha_i^\tau c_i q_s^\tau d\tau_0.$$

Directional differentiability of household spending gives

$$\begin{aligned} dc_{i,0}^{PP} &= \sum_{s=0}^{\infty} M_{i,0s} dy_{i,s}^{eq} \\ &= -\alpha_i^\tau c_i \left(\sum_{s=0}^{\infty} M_{i,0s} q_s^\tau \right) d\tau_0 \\ &= -\alpha_i^\tau c_i \Psi_i(q^\tau) d\tau_0. \end{aligned}$$

The bound in (34) guarantees absolute integrability, so aggregation and interchange of the household integral and time sum are valid. Thus

$$\begin{aligned} \frac{dC_0^{PP}}{C} &= \frac{1}{C} \int_{\mathcal{I}} dc_{i,0}^{PP} d\mu(i) \\ &= - \int_{\mathcal{I}} \Psi_i(q^\tau) \alpha_i^\tau \frac{c_i}{C} d\mu(i) d\tau_0 \\ &= -E^C[\Psi_i(q^\tau) \alpha_i^\tau] d\tau_0. \end{aligned}$$

Defining $s_\tau(q^\tau) = E^C[\Psi_i(q^\tau) \alpha_i^\tau]$ establishes the result. $\square$

Proof of Corollary 1. By the definition of covariance,

$$s_\tau(q^\tau) = \bar{\Psi}(q^\tau) \bar{\alpha}^\tau + \text{Cov}^C(\Psi_i(q^\tau), \alpha_i^\tau).$$

Consequently, the statistic equals the product-of-marginals expression based only on $\bar{\Psi}(q^\tau)$ and $\bar{\alpha}^\tau$ if and only if the covariance is zero. Relative to that benchmark, the normalized spending response

differs by

$$- \text{Cov}^C(\Psi_i(q^\tau), \alpha_i^\tau) d\tau_0.$$

For a positive tariff innovation under the nonnegative-exposure and nonnegative-response conditions stated above, a positive covariance makes the spending decline larger, while a negative covariance makes it smaller. $\square$

A.3 Proof of the between- and within-group decomposition

Let $\mathcal{B} = \sigma(b)$ be the sigma-algebra generated by the observable partition and define

$$m_\Psi(i) = E^C[\Psi_i(q^\tau) \mid \mathcal{B}] = \bar{\Psi}_{b(i)}(q^\tau), \quad m_\alpha(i) = E^C[\alpha_i^\tau \mid \mathcal{B}] = \bar{\alpha}_{b(i)}^\tau.$$

Because $\Psi_i(q^\tau) \in L^1(\mathbb{P}^C)$ and $\alpha_i^\tau \in L^\infty(\mathbb{P}^C)$, every term in the following calculation is integrable. Adding and subtracting $E^C[m_\Psi m_\alpha]$ gives

$$\begin{aligned} \text{Cov}^C(\Psi_i(q^\tau), \alpha_i^\tau) &= E^C[\Psi_i(q^\tau) \alpha_i^\tau] - E^C[m_\Psi m_\alpha] \\ &\quad + E^C[m_\Psi m_\alpha] - E^C[m_\Psi] E^C[m_\alpha] \\ &= E^C \left[\text{Cov}_{b(i)}^C(\Psi_i(q^\tau), \alpha_i^\tau) \right] + \text{Cov}^C(m_\Psi, m_\alpha). \end{aligned} \quad (36)$$

The first equality in the final line follows from the law of iterated expectations:

$$E^C[\Psi_i(q^\tau) \alpha_i^\tau \mid \mathcal{B}] - m_\Psi m_\alpha = \text{Cov}_{b(i)}^C(\Psi_i(q^\tau), \alpha_i^\tau).$$

Equation (36) is therefore the law of total covariance under the expenditure-weighted measure.

Proof of Proposition 2. By (36),

$$\text{Cov}^C(\Psi_i(q^\tau), \alpha_i^\tau) = V_\tau(q^\tau) + H_\tau(q^\tau).$$

Combining this identity with Proposition 1 and the product-of-marginals decomposition proves

$$\frac{dC_0^{PP}}{C} = - [\bar{\Psi}(q^\tau) \bar{\alpha}^\tau + V_\tau(q^\tau) + H_\tau(q^\tau)] d\tau_0.$$

It also shows that the aggregate product-of-marginals expression is exact if and only if $V_\tau(q^\tau) + H_\tau(q^\tau) = 0$. This condition permits cancellation between nonzero between- and within-group components.

The statistic recoverable from group expenditure shares and group means is

$$\begin{aligned} s_\tau^{\text{group}}(q^\tau) &= \sum_{b \in \mathcal{B}} \pi_b^C \bar{\Psi}_b(q^\tau) \bar{\alpha}_b^\tau \\ &= \bar{\Psi}(q^\tau) \bar{\alpha}^\tau + V_\tau(q^\tau). \end{aligned} \quad (37)$$

On the other hand, iterated expectations give

$$\begin{aligned} s_\tau(q^\tau) &= \sum_{b \in \mathcal{B}} \pi_b^C [\bar{\Psi}_b(q^\tau) \bar{\alpha}_b^\tau + \text{Cov}_b^C(\Psi_i(q^\tau), \alpha_i^\tau)] \\ &= s_\tau^{\text{group}}(q^\tau) + H_\tau(q^\tau). \end{aligned}$$

Hence group expenditure shares and group-level averages recover the exact statistic if and only if $H_\tau(q^\tau) = 0$. $\square$

Proof of Corollary 2. Index the two economies by $k \in \{1, 2\}$. By assumption, for every group with positive expenditure share,

$$\pi_{b,1}^C = \pi_{b,2}^C = \pi_b^C, \quad \bar{\Psi}_{b,1}(q^\tau) = \bar{\Psi}_{b,2}(q^\tau) = \bar{\Psi}_b(q^\tau), \quad \bar{\alpha}_{b,1}^\tau = \bar{\alpha}_{b,2}^\tau = \bar{\alpha}_b^\tau.$$

The common shares and conditional means imply common aggregate means by iterated expectations. Applying (37) in each economy yields

$$s_{\tau,k}(q^\tau) = \sum_{b \in \mathcal{B}} \pi_b^C \bar{\Psi}_b(q^\tau) \bar{\alpha}_b^\tau + H_{\tau,k}(q^\tau).$$

Therefore

$$s_{\tau,1}(q^\tau) - s_{\tau,2}(q^\tau) = H_{\tau,1}(q^\tau) - H_{\tau,2}(q^\tau), \quad (38)$$

and the difference between their normalized direct spending responses is

$$\frac{dC_{0,1}^{PP}}{C_1} - \frac{dC_{0,2}^{PP}}{C_2} = -[H_{\tau,1}(q^\tau) - H_{\tau,2}(q^\tau)] d\tau_0.$$

For a nonzero tariff innovation, distinct horizontal terms therefore imply distinct normalized direct aggregate spending responses. $\square$

B Full model and derivations

B.1 Prices, tariffs, and timing

Let $P_{H,t}$ denote the price of the domestic final good and let E_t denote the domestic-currency price of foreign currency. The foreign producer price of imports is P_t^* . Define the real exchange rate as

$$q_t \equiv \frac{E_t P_t^*}{P_{H,t}}. \quad (39)$$

An increase in q_t is a real depreciation.

The tariff-inclusive relative prices of imported consumption goods and imported intermediate inputs are

$$p_{M,t}^C = (1 + \tau_t^C) q_t, \quad p_{M,t}^I = (1 + \tau_t^I) q_t. \quad (40)$$

The border price is q_t . Domestic households and firms pay the tariff-inclusive prices, but foreign suppliers receive the border price.

Because the steady-state tariff can be zero, tariff shocks are written in terms of the log tariff wedge,

$$\vartheta_t^j \equiv \log(1 + \tau_t^j) - \log(1 + \bar{\tau}^j), \quad j \in \{C, I\}. \quad (41)$$

Each wedge follows

$$\vartheta_t^j = \rho_{\tau,j} \vartheta_{t-1}^j + \varepsilon_{\tau,j,t}. \quad (42)$$

The final-goods experiment sets $\varepsilon_{\tau,I,t} = 0$, the input-tariff experiment sets $\varepsilon_{\tau,C,t} = 0$, and the common tariff applies the same innovation to both wedges.

Domestic final-good inflation, nominal wage inflation, and CPI inflation are

$$\Pi_{H,t} \equiv \frac{P_{H,t}}{P_{H,t-1}}, \quad \Pi_t^W \equiv \frac{W_t}{W_{t-1}}, \quad \Pi_t^{CPI} \equiv \frac{P_t^{CPI}}{P_{t-1}^{CPI}}. \quad (43)$$

The real wage is

$$w_t \equiv \frac{W_t}{P_{H,t}}. \quad (44)$$

Consequently,

$$\Pi_{H,t} = \Pi_t^W \frac{w_{t-1}}{w_t}. \quad (45)$$

Households enter period t with liquid assets $a_{i,t-1}$. The fund return is realized, labor income and transfers are received, and the household chooses consumption and end-of-period assets $a_{i,t}$.

B.2 Households

Preferences and individual states. Households differ in liquid wealth a , idiosyncratic labor productivity e , a permanent discount factor β , and a permanent basket type ξ . The individual state at the beginning of period t is

$$z_{i,t} = (a_{i,t-1}, e_{i,t}, \beta_i, \xi_i). \quad (46)$$

Labor productivity follows a finite-state Markov chain,

$$\Pr(e_{i,t+1} = e' \mid e_{i,t} = e) = \Pi_e(e, e'). \quad (47)$$

The permanent types β_i and ξ_i do not change over time. Their joint population distribution is denoted by $\pi_{\beta,\xi}$. In the baseline,

$$\pi_{\beta,\xi} = \pi_\beta \pi_\xi. \quad (48)$$

All households work the common number of hours N_t , but their labor income is proportional

to productivity $e_{i,t}$. Period utility is

$$U_{i,t} = u(c_{i,t}) - v(N_t), \quad (49)$$

where

$$u(c) = \begin{cases} \frac{c^{1-\sigma} - 1}{1-\sigma}, & \sigma \neq 1, \\ \log c, & \sigma = 1, \end{cases} \quad (50)$$

and

$$v(N) = \chi_N \frac{N^{1+\varphi}}{1+\varphi}. \quad (51)$$

Intratemporal consumption problem. Household i consumes the domestic final good $c_{H,i,t}$ and the imported final good $c_{M,i,t}$. The consumption composite is

$$c_{i,t} = \left[(1 - \omega_{\xi_i})^{1/\eta_C} (c_{H,i,t} - \bar{c}_H)^{\frac{\eta_C - 1}{\eta_C}} + \omega_{\xi_i}^{1/\eta_C} (c_{M,i,t} - \bar{c}_M)^{\frac{\eta_C - 1}{\eta_C}} \right]^{\frac{\eta_C}{\eta_C - 1}}. \quad (52)$$

The household solves

$$\min_{c_{H,i,t}, c_{M,i,t}} \{c_{H,i,t} + p_{M,t}^C c_{M,i,t}\} \quad (53)$$

subject to equation (52) and

$$c_{H,i,t} \geq \bar{c}_H, \quad c_{M,i,t} \geq \bar{c}_M. \quad (54)$$

The demand system assumes $p_{M,t}^C > 0$, $\eta_C > 0$, and $\omega_{\xi} \in (0, 1)$; expressions at $\eta_C = 1$ are understood as the continuous Cobb–Douglas limit. Utility requires $c_{i,t} > 0$, and asset returns satisfy $1 + r_t^a > 0$. The closed-form conditional demands below apply on the interior, where both supernumerary quantities are positive. At the calibrated stationary equilibrium and for all reported first-order perturbations, implied levels remain positive and the household budget admits the borrowing-constraint choice. State-by-state strict feasibility requires

$$(1 + r_t^a) a_{i,t-1} + (1 - \tau_t^\ell) w_t e_{i,t} N_t + T_t(z) - \underline{a} > \zeta_t.$$

The usual no-Ponzi/transversality condition is also imposed. If a subsistence or feasibility inequality were to bind in levels, the interior demand formulas and their Jacobians would have to be replaced by the corresponding constrained problem.

Define supernumerary consumption by

$$\tilde{c}_{H,i,t} \equiv c_{H,i,t} - \bar{c}_H, \quad \tilde{c}_{M,i,t} \equiv c_{M,i,t} - \bar{c}_M. \quad (55)$$

The first-order condition for the relative allocation is

$$\frac{\tilde{c}_{M,i,t}}{\tilde{c}_{H,i,t}} = \frac{\omega_{\xi_i}}{1 - \omega_{\xi_i}} (p_{M,t}^C)^{-\eta_C}. \quad (56)$$

The supernumerary price index is

$$\mathcal{P}_{\xi,t} = [(1 - \omega_{\xi}) + \omega_{\xi}(p_{M,t}^C)^{1-\eta_C}]^{\frac{1}{1-\eta_C}}. \quad (57)$$

The cost of the subsistence bundle is

$$\zeta_t = \bar{c}_H + p_{M,t}^C \bar{c}_M. \quad (58)$$

Conditional demands are

$$c_{H,i,t} = \bar{c}_H + (1 - \omega_{\xi_i}) \mathcal{P}_{\xi_i,t}^{\eta_C} c_{i,t}, \quad (59)$$

$$c_{M,i,t} = \bar{c}_M + \omega_{\xi_i} \left(\frac{\mathcal{P}_{\xi_i,t}}{p_{M,t}^C} \right)^{\eta_C} c_{i,t}. \quad (60)$$

The minimum expenditure required to obtain composite consumption $c_{i,t}$ is

$$x_{i,t} = \zeta_t + \mathcal{P}_{\xi_i,t} c_{i,t}. \quad (61)$$

Define the supernumerary import expenditure share by

$$\vartheta_{\xi,t} \equiv \omega_{\xi} \left(\frac{p_{M,t}^C}{\mathcal{P}_{\xi,t}} \right)^{1-\eta_C}. \quad (62)$$

The household's total import expenditure share is

$$s_{M,i,t} = \frac{p_{M,t}^C \bar{c}_M + \vartheta_{\xi_i,t} \mathcal{P}_{\xi_i,t} c_{i,t}}{\zeta_t + \mathcal{P}_{\xi_i,t} c_{i,t}}. \quad (63)$$

Holding prices and basket type fixed,

$$\frac{\partial s_{M,i,t}}{\partial \log x_{i,t}} = \vartheta_{\xi_i,t} - s_{M,i,t}. \quad (64)$$

If imported goods are necessity-like, $s_{M,i,t} > \vartheta_{\xi_i,t}$, the import share falls with household expenditure.

Intertemporal problem. Let r_t^a denote the realized net return on the mutual-fund claim. Given the aggregate state, household i solves

$$V_t(a_-, e; \beta, \xi) = \max_{c, a'} u(c) - v(N_t) + \beta \sum_{e'} \Pi_e(e, e') V_{t+1}(a', e'; \beta, \xi) \quad (65)$$

$$\text{s.t.} \quad \zeta_t + \mathcal{P}_{\xi,t} c + a' = (1 + r_t^a) a_- + (1 - \tau_t^{\ell}) w_t e N_t + T_t(e, \beta, \xi), \quad (66)$$

$$a' \geq \underline{a}. \quad (67)$$

The Lagrangian is

$$\begin{aligned}\mathcal{L} = & u(c) - v(N_t) + \beta \sum_{e'} \Pi_e(e, e') V_{t+1}(a', e'; \beta, \xi) \\ & + \lambda \left[(1 + r_t^a) a_- + (1 - \tau_t^\ell) w_t e N_t + T_t - \zeta_t - \mathcal{P}_{\xi, t} c - a' \right] \\ & + \mu(a' - \underline{a}).\end{aligned}\tag{68}$$

The first-order condition for composite consumption is

$$u'(c_{i,t}) = \lambda_{i,t} \mathcal{P}_{\xi_{i,t}}.\tag{69}$$

Hence, the marginal value of one unit of domestic-good resources is

$$\lambda_{i,t} = \frac{u'(c_{i,t})}{\mathcal{P}_{\xi_{i,t}}}.\tag{70}$$

The first-order condition for end-of-period assets is

$$\lambda_{i,t} = \beta_i \sum_{e'} \Pi_e(e_{i,t}, e') V_{a,t+1}(a_{i,t}, e'; \beta_i, \xi_i) + \mu_{i,t}.\tag{71}$$

The envelope condition is

$$V_{a,t}(a_{i,t-1}, e_{i,t}; \beta_i, \xi_i) = (1 + r_t^a) \lambda_{i,t}.\tag{72}$$

Combining the two conditions gives the Euler inequality

$$\lambda_{i,t} \geq \beta_i \sum_{e'} \Pi_e(e_{i,t}, e') (1 + r_{t+1}^a) \lambda_{i,t+1}(e'),\tag{73}$$

with equality whenever $a_{i,t} > \underline{a}$. The complementary slackness conditions are

$$\mu_{i,t} \geq 0, \quad a_{i,t} - \underline{a} \geq 0, \quad \mu_{i,t}(a_{i,t} - \underline{a}) = 0.\tag{74}$$

For CRRA utility, the interior Euler equation implies

$$c_{i,t} = \left[\mathcal{P}_{\xi_{i,t}} \beta_i \sum_{e'} \Pi_e(e_{i,t}, e') (1 + r_{t+1}^a) \lambda_{i,t+1}(e') \right]^{-1/\sigma}.\tag{75}$$

Given $a_{i,t}$ and the implied $c_{i,t}$, the endogenous current asset position is

$$a_{i,t-1} = \frac{\zeta_t + \mathcal{P}_{\xi_{i,t}} c_{i,t} + a_{i,t} - (1 - \tau_t^\ell) w_t e_{i,t} N_t - T_t(e_{i,t}, \beta_i, \xi_i)}{1 + r_t^a}.\tag{76}$$

Equations (75)–(76) define the endogenous-grid step used in the numerical solution.

Distribution. Let

$$\mathcal{D}_t(a_-, e, \beta, \xi) \quad (77)$$

denote the beginning-of-period distribution. Given the asset policy $a'_t = g_{a,t}(a_-, e, \beta, \xi)$, the distribution evolves according to

$$\begin{aligned} \mathcal{D}_{t+1}(a', e', \beta, \xi) &= \sum_e \int \mathbf{1}\{g_{a,t}(a_-, e, \beta, \xi) \leq a'\} \\ &\quad \times \Pi_e(e, e') d\mathcal{D}_t(a_-, e, \beta, \xi). \end{aligned} \quad (78)$$

In the discrete implementation, the asset policy is lottery-interpolated onto the fixed asset grid, which preserves total probability mass.

Aggregation. Aggregate end-of-period household assets are

$$A_t = \int g_{a,t}(z) d\mathcal{D}_t(z). \quad (79)$$

Aggregate composite consumption is

$$C_t = \int c_t(z) d\mathcal{D}_t(z). \quad (80)$$

Aggregate household expenditure, denoted by $\mathcal{S}_t$ to distinguish it unambiguously from domestic value added X_t , is

$$\mathcal{S}_t = \int x_t(z) d\mathcal{D}_t(z). \quad (81)$$

Aggregate domestic and imported final consumption are

$$C_{H,t} = \int c_{H,t}(z) d\mathcal{D}_t(z), \quad (82)$$

$$C_{M,t} = \int c_{M,t}(z) d\mathcal{D}_t(z). \quad (83)$$

The expenditure identity is

$$\mathcal{S}_t = C_{H,t} + p_{M,t}^C C_{M,t}. \quad (84)$$

Normalize mean labor productivity to one:

$$\int e d\mathcal{D}_t(z) = 1. \quad (85)$$

Aggregate labor income is therefore

$$w_t N_t. \quad (86)$$

The productivity-weighted marginal value of labor income is

$$\Lambda_t^e = \int e\lambda_t(z) d\mathcal{D}_t(z). \quad (87)$$

This is the household object entering the wage-setting block.

B.3 Tariff exposure inside the household problem

Differentiate household expenditure with respect to the relative import price while holding composite consumption fixed:

$$\frac{\partial x_{i,t}}{\partial p_{M,t}^C} = \bar{c}_M + \frac{\partial \mathcal{P}_{\xi,t}}{\partial p_{M,t}^C} c_{i,t}. \quad (88)$$

The supernumerary price index satisfies

$$\frac{\partial \mathcal{P}_{\xi,t}}{\partial p_{M,t}^C} = \vartheta_{\xi,t} \frac{\mathcal{P}_{\xi,t}}{p_{M,t}^C}. \quad (89)$$

It follows that

$$\frac{\partial \log x_{i,t}}{\partial \log p_{M,t}^C} = s_{M,i,t}. \quad (90)$$

Since

$$p_{M,t}^C = (1 + \tau_t^C) q_t, \quad (91)$$

the exposure to the primitive log tariff wedge $\vartheta_t^C = \log(1 + \tau_t^C) - \log(1 + \bar{\tau}^C)$, holding the border price fixed, is

$$\alpha_{i,t}^\tau \equiv \frac{\partial \log x_{i,t}}{\partial \vartheta_t^C} = s_{M,i,t}. \quad (92)$$

If the primitive were instead the ad valorem rate itself, the derivative would be $s_{M,i,t}/(1 + \tau_t^C)$. More generally, allowing for incomplete pass-through,

$$\alpha_{i,t}^\tau = \kappa_{\tau,t} s_{M,i,t}, \quad \kappa_{\tau,t} \equiv \frac{\partial \log p_{M,t}^C}{\partial \vartheta_t^C}. \quad (93)$$

B.4 The partial-equilibrium statistic inside HANK

Let $\bar{z}$ denote a household state in the stationary distribution. Define

$$M_{\bar{z},0s} \equiv \left. \frac{\partial x_{\bar{z},0}}{\partial y_{\bar{z},s}} \right|_{ss} \quad (94)$$

as the date-0 expenditure response to one unit of liquid resources received at date s .

Let q_s^τ denote the exogenous tariff-wedge path per unit innovation, normalized by $q_0^\tau = 1$. This object excludes the endogenous border-price response, which belongs to the general-equilibrium

experiment. Under the AR(1) process, $q_s^\tau = \rho_\tau^s$. The tariff-relevant iMPC is

$$\Psi_{\bar{z}}(q^\tau) = \sum_{s=0}^{\infty} M_{\bar{z},0s} q_s^\tau. \quad (95)$$

Holding the distribution, labor income, transfers, asset returns, and all non-import prices fixed, the tariff generates the resource-equivalent loss

$$dy_{\bar{z},s}^{\text{eq}} = -\alpha_{\bar{z}}^\tau \bar{x}_{\bar{z}} q_s^\tau d\vartheta_0^C. \quad (96)$$

The date-0 expenditure response of household $\bar{z}$ is

$$\begin{aligned} dx_{\bar{z},0}^{PP} &= \sum_{s=0}^{\infty} M_{\bar{z},0s} dy_{\bar{z},s}^{\text{eq}} \\ &= -\Psi_{\bar{z}}(q^\tau) \alpha_{\bar{z}}^\tau \bar{x}_{\bar{z}} d\vartheta_0^C. \end{aligned} \quad (97)$$

The quantitative implementation constructs the complete state-by-date matrix $M_{\bar{z},0s}$ from the same fake-news recursion that underlies the aggregate sequence-space Jacobian. The benchmark uses 140 quarterly columns and $q_s^\tau = \rho_\tau^s$. Aggregating each column over stationary states reproduces the corresponding row of the aggregate expenditure–transfer Jacobian. The main text reports both the path-weighted statistic at $\rho_\tau = 0.75$ and the contemporaneous special case $q_0^\tau = 1$, $q_s^\tau = 0$ for $s > 0$. Horizon and finite-difference checks are reported in the replication package.

Figure 6 makes the timing distinction explicit. The familiar cumulative MPC traces responses after an impact transfer, while the tariff-relevant statistic uses the date-0 response to transfers dated along the tariff path.

Define the expenditure-weighted stationary expectation as

$$\mathbb{E}^S[h_{\bar{z}}] = \int h_{\bar{z}} \frac{\bar{x}_{\bar{z}}}{\bar{S}} d\bar{D}(\bar{z}). \quad (98)$$

Aggregating equation (97) gives

$$\frac{dS_0^{PP}}{\bar{S}} = -\mathbb{E}^S[\Psi_{\bar{z}}(q^\tau) \alpha_{\bar{z}}^\tau] d\vartheta_0^C. \quad (99)$$

For any observable partition $b(\bar{z})$,

$$\mathbb{E}^S[\Psi_{\bar{z}} \alpha_{\bar{z}}^\tau] = \bar{\Psi} \bar{\alpha}^\tau + V_\tau + H_\tau, \quad (100)$$

where

$$\bar{\Psi} = \mathbb{E}^S[\Psi_{\bar{z}}], \quad \bar{\alpha}^\tau = \mathbb{E}^S[\alpha_{\bar{z}}^\tau], \quad (101)$$

$$V_\tau = \text{Cov}^S(\mathbb{E}^S[\Psi_{\bar{z}} | b(\bar{z})], \mathbb{E}^S[\alpha_{\bar{z}}^\tau | b(\bar{z})]), \quad (102)$$

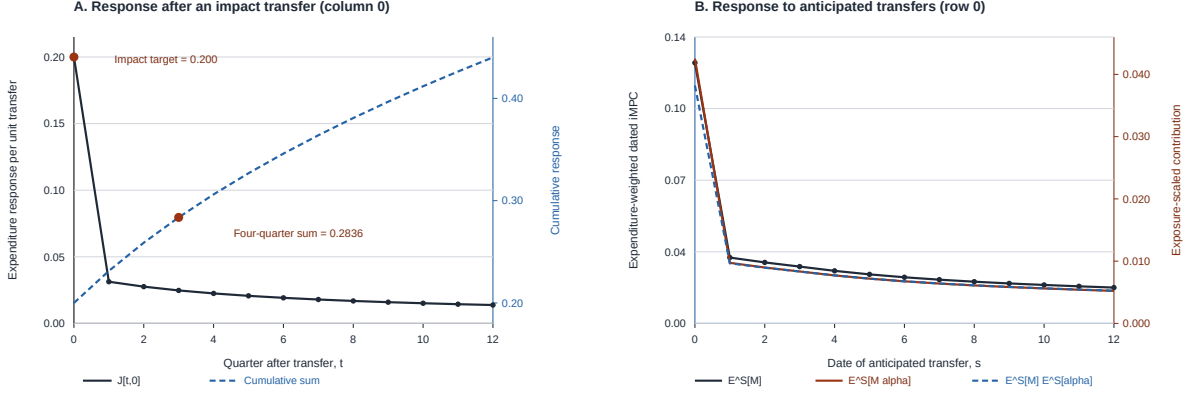


Figure 6: Dated spending responses: post-transfer dynamics and anticipated-transfer exposure

Notes: Panel A reports the population-average expenditure response at date t to a one-unit liquid transfer at date 0, $J_{t,0}^{S,T}$, together with its cumulative sum. The impact response is the targeted quarterly MPC of 0.200; the four-quarter cumulative response of 0.2836 is an untargeted model implication. Panel B reports the expenditure-weighted date-0 response to a transfer anticipated for date s , $E^S[M_{z,0s}]$, together with the exposure-weighted joint term and the product of marginal means. Panel A traces column 0 of the aggregate expenditure-transfer Jacobian. Panel B aggregates the state-level row-0 derivatives using expenditure weights. These objects are not interchangeable: only Panel B enters $\Psi_z(q^\tau) = \sum_s M_{z,0s} q_s^\tau$. The right axis in Panel B reports exposure-scaled contributions; expenditure-weighted mean exposure is 0.300. Dates are quarters; calculations use the validated 140-quarter household Jacobians, with quarters 0–12 displayed.

and

$$H_\tau = \mathbb{E}^S [\text{Cov}^S(\Psi_{\bar{z}}, \alpha_{\bar{z}}^\tau \mid b(\bar{z}))]. \quad (103)$$

Both components are defined relative to the chosen observable partition. In particular, H_τ is a within-group covariance residual and should not be interpreted as the incremental equilibrium effect of permanent basket-type dispersion, which is measured by the vertical–full model comparison.

Equation (99) is a partial-equilibrium derivative evaluated inside the HANK household block. The full general-equilibrium response additionally allows wages, hours, taxes, transfers, returns, relative prices, and the distribution to adjust.

B.5 Production and imported inputs

Domestic value added. Domestic producers use effective labor to produce the domestic value-added input:

$$X_t = Z_t N_t, \quad (104)$$

where X_t is domestic value added and Z_t is aggregate productivity.

Let $P_{X,t}$ denote the nominal price of domestic value added and define

$$p_{X,t} \equiv \frac{P_{X,t}}{P_{H,t}}. \quad (105)$$

Value added is supplied competitively, so its relative unit cost is

$$p_{X,t} = \frac{w_t}{Z_t}. \quad (106)$$

Gross-output producers. Gross-output producers combine domestic value added $X_{H,t}$ and imported inputs $M_{I,t}$:

$$Y_t = \Gamma_Y \left[(1 - \alpha_I)^{1/\nu_I} X_{H,t}^{\frac{\nu_I-1}{\nu_I}} + \alpha_I^{1/\nu_I} M_{I,t}^{\frac{\nu_I-1}{\nu_I}} \right]^{\frac{\nu_I}{\nu_I-1}}. \quad (107)$$

Let $c_{Y,t}$ denote the minimum unit cost of the unscaled CES composite:

$$c_{Y,t} = \left[(1 - \alpha_I) p_{X,t}^{1-\nu_I} + \alpha_I (p_{M,t}^I)^{1-\nu_I} \right]^{\frac{1}{1-\nu_I}}. \quad (108)$$

With flexible prices, producers charge the constant markup

$$\mu_p = \frac{\epsilon_p}{\epsilon_p - 1} \quad (109)$$

over the scale-adjusted unit cost $c_{Y,t}/\Gamma_Y$. Because $P_{H,t}$ is the gross-output price,

$$1 = \mu_p \frac{c_{Y,t}}{\Gamma_Y}. \quad (110)$$

The first-order conditions imply

$$X_{H,t} = (1 - \alpha_I) \left(\frac{p_{X,t}}{c_{Y,t}} \right)^{-\nu_I} \frac{Y_t}{\Gamma_Y}, \quad (111)$$

$$M_{I,t} = \alpha_I \left(\frac{p_{M,t}^I}{c_{Y,t}} \right)^{-\nu_I} \frac{Y_t}{\Gamma_Y}. \quad (112)$$

Real dividends are denoted by Div_t , distinct from the household distribution $\mathcal{D}_t$. Consolidating domestic production, they equal gross-output revenue net of labor costs, imported-input costs, and the fixed operating cost:

$$\text{Div}_t = Y_t - w_t N_t - p_{M,t}^I M_{I,t} - F_t. \quad (113)$$

The operating cost is measured in units of the domestic final good and is held fixed in levels along the transition, $F_t = \bar{F}$. Since variable costs equal $(c_{Y,t}/\Gamma_Y)Y_t = Y_t/\mu_p$,

$$\text{Div}_t = \left(1 - \frac{1}{\mu_p} \right) Y_t - F_t. \quad (114)$$

The domestic value-added market clears:

$$X_{H,t} = X_t. \quad (115)$$

Equations (104), (111), and (115) determine the link between GDP, gross output, and labor.

Imported inputs are part of gross output but not domestic value added. They therefore do not enter GDP directly.

Exports. Foreign demand for the domestic final good is

$$EX_t = \overline{EX} q_t^{\eta_X}, \quad (116)$$

where $\eta_X > 0$. A real depreciation increases exports.

B.6 Wage setting

There is a continuum of differentiated labor services indexed by $j \in [0, 1]$. Aggregate labor is

$$N_t = \left[\int_0^1 N_t(j)^{\frac{\epsilon_W - 1}{\epsilon_W}} dj \right]^{\frac{\epsilon_W}{\epsilon_W - 1}}. \quad (117)$$

Labor demand for variety j is

$$N_t(j) = \left(\frac{W_t(j)}{W_t} \right)^{-\epsilon_W} N_t. \quad (118)$$

The aggregate wage index satisfies

$$W_t^{1-\epsilon_W} = \int_0^1 W_t(j)^{1-\epsilon_W} dj. \quad (119)$$

Each union can reset its nominal wage with probability $1 - \theta_W$. A resetting union chooses W_t^* , knowing that this wage remains in place with probability θ_W^k at date $t + k$. Its labor demand at that date is

$$N_{t+k|t} = \left(\frac{W_t^*}{W_{t+k}} \right)^{-\epsilon_W} N_{t+k}. \quad (120)$$

The marginal utility benefit from one additional unit of real labor income is $(1 - \tau_t^\ell) \Lambda_t^e$. The reset-wage first-order condition is

$$0 = \sum_{k=0}^{\infty} (\bar{\beta}_W \theta_W)^k \mathbb{E}_t \left[N_{t+k|t} \left\{ (\epsilon_W - 1)(1 - \tau_{t+k}^\ell) \Lambda_{t+k}^e \frac{W_t^*}{P_{H,t+k}} - \epsilon_W v'(N_{t+k|t}) \right\} \right], \quad (121)$$

where $\bar{\beta}_W$ is the discount factor used by the wage-setting union.

The desired gross wage markup is

$$\mu_W = \frac{\epsilon_W}{\epsilon_W - 1}. \quad (122)$$

Under flexible wages, the real wage satisfies

$$w_t^{\text{flex}} = \mu_W \frac{v'(N_t)}{(1 - \tau_t^\ell) \Lambda_t^\epsilon}. \quad (123)$$

The Calvo wage index evolves according to

$$W_t^{1-\epsilon_W} = \theta_W W_{t-1}^{1-\epsilon_W} + (1 - \theta_W) (W_t^*)^{1-\epsilon_W}. \quad (124)$$

Define the wage-markup gap

$$\hat{\mu}_t^W \equiv \log \left(\frac{w_t^{\text{flex}}}{w_t} \right). \quad (125)$$

A first-order approximation around the zero-inflation stationary equilibrium gives

$$\pi_t^W = \bar{\beta}_W \mathbb{E}_t \pi_{t+1}^W + \kappa_W \hat{\mu}_t^W, \quad (126)$$

where

$$\kappa_W = \frac{(1 - \theta_W)(1 - \bar{\beta}_W \theta_W)}{\theta_W(1 + \epsilon_W \varphi)}. \quad (127)$$

Wage dispersion is second order around the symmetric stationary equilibrium and does not enter the first-order transition system.

B.7 Household cost-of-living indices and CPI

For a household with stationary composite consumption $\bar{c}_z$, define the compensated cost-of-living index as

$$P_{z,t}^{COL} = P_{H,t} \frac{\zeta_t + \mathcal{P}_{\xi,t} \bar{c}_z}{\bar{\zeta} + \bar{\mathcal{P}}_{\xi} \bar{c}_z}. \quad (128)$$

This index measures the expenditure required at date t to purchase the household's stationary utility level relative to its stationary expenditure.

Define stationary expenditure weights by

$$\omega_z^S = \frac{\bar{x}_z d\bar{\mathcal{D}}(z)}{\bar{\mathcal{S}}}. \quad (129)$$

The aggregate compensated CPI is

$$\log P_t^{CPI} = \int \log P_{z,t}^{COL} \omega_z^S. \quad (130)$$

To first order,

$$d \log P_t^{CPI} = d \log P_{H,t} + \bar{s}_M d \log p_{M,t}^C, \quad (131)$$

where

$$\bar{s}_M = \frac{\bar{p}_M^C \bar{C}_M}{\bar{\mathcal{S}}} \quad (132)$$

is the stationary final-import expenditure share.

CPI inflation therefore satisfies

$$\pi_t^{CPI} = (1 - \bar{s}_M)\pi_{H,t} + \bar{s}_M\pi_{M,t}^C, \quad (133)$$

where

$$\pi_{M,t}^C = \pi_{H,t} + \Delta \log p_{M,t}^C. \quad (134)$$

B.8 Monetary policy and asset prices

Monetary policy. The central bank follows

$$\begin{aligned} \log \left(\frac{1 + i_t}{1 + \bar{i}} \right) &= \rho_i \log \left(\frac{1 + i_{t-1}}{1 + \bar{i}} \right) \\ &+ (1 - \rho_i) \left[\phi_\pi \log \Pi_t^{CPI} + \phi_X \log \left(\frac{X_t}{\bar{X}} \right) \right]. \end{aligned} \quad (135)$$

The ex ante domestic real gross rate is

$$R_t = \frac{1 + i_t}{\mathbb{E}_t \Pi_{H,t+1}}. \quad (136)$$

The realized real return on nominal government debt issued at $t - 1$ is

$$R_t^B = \frac{1 + i_{t-1}}{\Pi_{H,t}}. \quad (137)$$

Uncovered interest parity. Let R_t^* denote the foreign real gross interest rate. Perfect capital mobility implies

$$R_t = R_t^* \mathbb{E}_t \left(\frac{q_{t+1}}{q_t} \right). \quad (138)$$

In log-linear form,

$$\hat{R}_t - \hat{R}_t^* = \mathbb{E}_t \Delta \hat{q}_{t+1}. \quad (139)$$

The baseline sets $\phi_{NFA} = 0$. The NFA-premium robustness modifies only the first-order condition:

$$\hat{R}_t - \hat{R}_t^* = \mathbb{E}_t \Delta \hat{q}_{t+1} - \phi_{NFA} dNFA_t, \quad (140)$$

where $dNFA_t$ is the level deviation normalized by stationary domestic value added, $\bar{X}$. Thus equation (139) is recovered under the baseline calibration.

The realized return on foreign assets in domestic-good units is

$$R_t^F = R_{t-1}^* \frac{q_t}{q_{t-1}}. \quad (141)$$

B.9 Mutual fund

Households hold a single liquid claim issued by a competitive mutual fund. At the end of period t , the fund holds domestic equity J_t , government debt B_t , and net foreign assets NFA_t :

$$A_t = J_t + B_t + NFA_t. \quad (142)$$

Domestic equity is a claim to future dividends. In a deterministic transition, its value satisfies

$$J_t = \frac{\text{Div}_{t+1} + J_{t+1}}{R_t}. \quad (143)$$

More generally,

$$J_t = \mathbb{E}_t \left[\frac{\text{Div}_{t+1} + J_{t+1}}{R_t} \right]. \quad (144)$$

The realized payoff on the fund's beginning-of-period portfolio is

$$(1 + r_t^a)A_{t-1} = \text{Div}_t + J_t + R_t^B B_{t-1} + R_t^F NFA_{t-1}. \quad (145)$$

Thus,

$$1 + r_t^a = \frac{\text{Div}_t + J_t + R_t^B B_{t-1} + R_t^F NFA_{t-1}}{A_{t-1}}. \quad (146)$$

This return includes capital gains and losses on domestic equity. In particular, the period-zero household return incorporates the impact revaluation of J_t .

B.10 Fiscal policy

Tariff revenue is

$$\mathcal{R}_t^\tau = \tau_t^C q_t C_{M,t} + \tau_t^I q_t M_{I,t}. \quad (147)$$

Imports are valued at the border price because the tariff wedge is domestic revenue.

The government budget constraint is

$$B_t = R_t^B B_{t-1} + G_t + \mathcal{T}_t - \tau_t^\ell w_t N_t - \mathcal{R}_t^\tau, \quad (148)$$

where

$$\mathcal{T}_t = \int T_t(z) d\mathcal{D}_t(z) \quad (149)$$

is aggregate transfers.

In the baseline,

$$\mathcal{T}_t = 0, \quad G_t = \bar{G}. \quad (150)$$

Tariff revenue therefore reduces government borrowing on impact.

The labor-tax rule is

$$\tau_t^\ell - \bar{\tau}^\ell = \psi_B \frac{B_{t-1} - \bar{B}}{\bar{X}}, \quad \psi_B > 0. \quad (151)$$

A small ψ_B implies gradual debt stabilization.

Under the uniform-rebate robustness,

$$T_t(z) = \mathcal{R}_t^\tau \quad \text{for all } z, \quad (152)$$

after adjusting for unit population mass.

B.11 External balance

The trade balance in domestic-final-good units is

$$TB_t = EX_t - q_t (C_{M,t} + M_{I,t}). \quad (153)$$

Tariff-inclusive import prices do not enter the foreign import bill.

Net foreign assets evolve according to

$$NFA_t = R_t^F NFA_{t-1} + TB_t. \quad (154)$$

Equivalently, the current account is

$$CA_t \equiv NFA_t - NFA_{t-1} = (R_t^F - 1)NFA_{t-1} + TB_t. \quad (155)$$

B.12 Market clearing

The domestic final-good resource constraint is

$$Y_t = C_{H,t} + G_t + EX_t + F_t. \quad (156)$$

The fixed operating cost is a real use of the domestic final good.

The domestic value-added market clears according to

$$X_t = X_{H,t}. \quad (157)$$

The labor market clears according to

$$X_t = Z_t N_t. \quad (158)$$

The asset market clears according to

$$A_t = J_t + B_t + NFA_t. \quad (159)$$

Combining household, firm, government, fund, and external budget constraints gives Walras' law. One aggregate market-clearing condition is therefore redundant once all remaining equilibrium conditions hold.

B.13 Equilibrium definition

Given the initial household distribution $\mathcal{D}_0$, initial asset positions, exogenous foreign variables, productivity, and tariff processes, an equilibrium is a sequence

$$\left\{ \begin{array}{c} c_t(z), x_t(z), a'_t(z), c_{H,t}(z), c_{M,t}(z), \mathcal{D}_t(z), \\ C_t, \mathcal{S}_t, C_{H,t}, C_{M,t}, N_t, X_t, Y_t, X_{H,t}, M_{I,t}, EX_t, \\ w_t, p_{X,t}, p_{M,t}^C, p_{M,t}^I, q_t, \\ \Pi_{H,t}, \Pi_t^W, \Pi_t^{CPI}, i_t, R_t, r_t^a, \\ \text{Div}_t, J_t, A_t, B_t, NFA_t, \tau_t^\ell, \mathcal{T}_t, \mathcal{R}_t^\tau \end{array} \right\}_{t \geq 0} \quad (160)$$

such that:

1. household expenditure and saving policies solve equations (65)–(67), with conditional consumption demands given by equations (59) and (60);
2. the household distribution $\mathcal{D}_t$ evolves according to equation (78), and household aggregates satisfy equations (79)–(83);
3. value-added producers and final-good firms satisfy equations (104)–(115), while exports satisfy equation (116);
4. wage setters and the aggregate wage index satisfy equations (121) and (124);
5. household cost-of-living indices and aggregate CPI inflation satisfy equations (128)–(133), and monetary policy satisfies equation (135);
6. the mutual fund and household asset return satisfy equations (142), (145), and (146);
7. the government satisfies tariff revenue, budget, and labor-tax equations (147), (148), and (151);
8. uncovered interest parity and external positions satisfy equations (138), (153)–(155);
9. domestic goods, value added, labor, and assets clear according to equations (156)–(159).

B.14 Stationary equilibrium

A stationary equilibrium has constant real quantities and zero domestic and wage inflation:

$$\bar{\Pi}_H = \bar{\Pi}^W = \bar{\Pi}^{CPI} = 1. \quad (161)$$

The stationary real rate satisfies

$$\bar{R} = 1 + \bar{i}. \quad (162)$$

The real exchange rate is normalized to

$$\bar{q} = 1. \quad (163)$$

For each permanent type (β, ξ) , stationary household policies solve

$$\bar{V}(a, e; \beta, \xi) = \max_{c, a'} u(c) - v(\bar{N}) + \beta \sum_{e'} \Pi_e(e, e') \bar{V}(a', e'; \beta, \xi) \quad (164)$$

$$\text{s.t. } \bar{\zeta} + \bar{\mathcal{P}}_\xi c + a' = (1 + \bar{r}^a)a + (1 - \bar{\tau}^\ell) \bar{w} e \bar{N} + \bar{T}(e, \beta, \xi). \quad (165)$$

The stationary distribution satisfies

$$\bar{\mathcal{D}} = \mathcal{T}(\bar{g}_a, \Pi_e) \bar{\mathcal{D}}, \quad (166)$$

where $\mathcal{T}$ is the distribution-transition operator.

The stationary production block satisfies

$$\bar{X} = \bar{Z} \bar{N}, \quad (167)$$

$$\bar{p}_X = \frac{\bar{w}}{\bar{Z}}, \quad (168)$$

$$\bar{Y} = \Gamma_Y \left[(1 - \alpha_I)^{1/\nu_I} \bar{X}^{\frac{\nu_I-1}{\nu_I}} + \alpha_I^{1/\nu_I} \bar{M}_I^{\frac{\nu_I-1}{\nu_I}} \right]^{\frac{\nu_I}{\nu_I-1}}, \quad (169)$$

$$\bar{c}_Y = \left[(1 - \alpha_I) \bar{p}_X^{1-\nu_I} + \alpha_I (\bar{p}_M^I)^{1-\nu_I} \right]^{\frac{1}{1-\nu_I}}, \quad (170)$$

$$1 = \mu_p \frac{\bar{c}_Y}{\Gamma_Y}, \quad (171)$$

$$\bar{X} = (1 - \alpha_I) \left(\frac{\bar{p}_X}{\bar{c}_Y} \right)^{-\nu_I} \frac{\bar{Y}}{\Gamma_Y}, \quad (172)$$

$$\bar{M}_I = \alpha_I \left(\frac{\bar{p}_M^I}{\bar{c}_Y} \right)^{-\nu_I} \frac{\bar{Y}}{\Gamma_Y}. \quad (173)$$

The stationary cost share and gross-output normalization imply $\alpha_I = 0.205784$ and $\Gamma_Y = 1.002157$ in the benchmark. Because Γ_Y is fixed, it does not enter the log-linear equations below.

The stationary fiscal and external positions satisfy

$$\bar{B} = \bar{R} \bar{B} + \bar{G} + \bar{T} - \bar{\tau}^\ell \bar{w} \bar{N} - \bar{\mathcal{R}}^\tau, \quad (174)$$

$$(1 - \bar{R}^F) \overline{NFA} = \overline{TB}. \quad (175)$$

The quantitative baseline sets

$$\frac{\bar{B}}{\bar{X}} = 2.32, \quad \overline{NFA} = 0, \quad \overline{TB} = 0. \quad (176)$$

Because the model is quarterly, $\bar{B}/\bar{X} = 2.32$ corresponds to 0.58 relative to annualized value added. With household financial wealth $\bar{A}/\bar{X} = 10$, the stationary mutual-fund balance sheet therefore implies $\bar{J}/\bar{X} = 7.68$. With zero stationary transfers and tariff revenue, the government budget reduces to $\bar{\tau}^\ell \bar{w} \bar{N} = \bar{G} + (\bar{R}^B - 1) \bar{B}$.

The fixed operating cost is chosen consistently with stationary dividends and the domestic

resource constraint:

$$\bar{F} = \bar{Y} - \bar{C}_H - \bar{G} - \bar{E}\bar{X}. \quad (177)$$

Equivalently,

$$\bar{F} = \left(1 - \frac{1}{\mu_p}\right) \bar{Y} - \bar{\text{Div}}. \quad (178)$$

The two expressions must coincide. In the benchmark, $\bar{F}/\bar{X} = 0.167$. This value is an accounting residual that reconciles the markup, household financial wealth, equity, and dividends. Because the same macroeconomic stationary equilibrium and fixed-cost level are used in the common-, vertical-, and full-exposure economies, it cannot mechanically generate differences across them; absolute responses nevertheless remain conditional on this stationary accounting calibration.

B.15 The three exposure economies

The quantitative analysis compares three household demand systems.

Common-basket economy. Let $\bar{\alpha}$ denote the aggregate stationary final-import expenditure share and let $\bar{\zeta} \equiv \bar{c}_H + \bar{c}_M$ at the normalized stationary import price. The common-basket economy removes both sources of exposure heterogeneity while preserving the stationary intertemporal problem:

$$\bar{c}_H^{\text{common}} = (1 - \bar{\alpha})\bar{\zeta}, \quad \bar{c}_M^{\text{common}} = \bar{\alpha}\bar{\zeta}, \quad \omega_\xi = \bar{\alpha} \quad \text{for all } \xi. \quad (179)$$

At the normalized stationary import price, every household has the same import expenditure share $\bar{\alpha}$.

Vertical-exposure economy. The vertical economy retains the Stone–Geary subsistence quantities but removes basket-type dispersion:

$$\bar{c}_H > 0, \quad \bar{c}_M > 0, \quad \omega_\xi = \omega^{\text{vertical}} \quad \text{for all } \xi. \quad (180)$$

Import shares vary with household expenditure but not with basket type conditional on expenditure.

Full-exposure economy. The full economy retains both non-homotheticity and basket-type heterogeneity:

$$\bar{c}_H > 0, \quad \bar{c}_M > 0, \quad \omega_\xi \in \{\omega_1, \dots, \omega_{N_\xi}\}. \quad (181)$$

The distribution of ω_ξ generates residual exposure dispersion conditional on expenditure. Its scale is a benchmark parameter assignment rather than a documented empirical target.

The aggregate final import share is matched across the three economies. Each household system is solved separately. In the baseline frictionless steady state, household policies and stationary distributions coincide numerically across systems when restricted to composite-consumption/saving objects and the marginal distribution over intertemporal states; the full system retains its independent basket-type label. The household Jacobians differ.

To see the stationary equivalence, impose the baseline independence of basket type from patience

and productivity. The normalized stationary import price implies $\mathcal{P}_\xi = 1$ in all three systems, and their total subsistence cost is the common value $\bar{\zeta}$. Hence every household faces the same stationary budget constraint $\bar{\zeta} + c + a' = \text{resources}$, with the same income process, discount factor, return, and borrowing constraint. Composite consumption and saving policies, and the marginal stationary distribution over intertemporal states, therefore coincide. The full system still carries a basket-type label. Conditional domestic and import demands, and especially their derivatives with respect to import prices, need not coincide; those derivatives are precisely what distinguishes the three sequence-space Jacobians.

B.16 First-order aggregate system

Let a hat denote a log deviation from stationary equilibrium. The log-linear unit-cost condition is

$$0 = (1 - \bar{s}_I)\widehat{p}_{X,t} + \bar{s}_I\widehat{p}_{M,t}^I, \quad (182)$$

where

$$\bar{s}_I = \frac{\bar{p}_M^I \bar{M}_I}{\bar{Y}/\mu_p}. \quad (183)$$

Flexible product pricing implies

$$\widehat{p}_{X,t} = \widehat{w}_t - \widehat{Z}_t. \quad (184)$$

Value added satisfies

$$\widehat{X}_t = \widehat{Z}_t + \widehat{N}_t. \quad (185)$$

Conditional input demands satisfy

$$\widehat{X}_t = \widehat{Y}_t - \nu_I \widehat{p}_{X,t}, \quad (186)$$

$$\widehat{M}_{I,t} = \widehat{Y}_t - \nu_I \widehat{p}_{M,t}^I. \quad (187)$$

Import prices satisfy

$$\widehat{p}_{M,t}^C = \widehat{q}_t + \vartheta_t^C, \quad (188)$$

$$\widehat{p}_{M,t}^I = \widehat{q}_t + \vartheta_t^I. \quad (189)$$

Exports satisfy

$$\widehat{EX}_t = \eta_X \widehat{q}_t. \quad (190)$$

The wage Phillips curve is

$$\pi_t^W = \bar{\beta}_W \mathbb{E}_t \pi_{t+1}^W + \kappa_W \left(\widehat{w}_t^{\text{flex}} - \widehat{w}_t \right). \quad (191)$$

The flexible real wage satisfies

$$\widehat{w}_t^{\text{flex}} = \varphi \widehat{N}_t - \widehat{\Lambda}_t^e + \frac{\bar{\tau}^\ell}{1 - \bar{\tau}^\ell} \widehat{\tau}_t^\ell. \quad (192)$$

Domestic inflation is

$$\pi_{H,t} = \pi_t^W - \Delta \widehat{w}_t. \quad (193)$$

CPI inflation is

$$\pi_t^{CPI} = \pi_{H,t} + \bar{s}_M \Delta \widehat{p}_{M,t}^C. \quad (194)$$

The log-linear Taylor rule is

$$\widehat{i}_t = \rho_i \widehat{i}_{t-1} + (1 - \rho_i) \left(\phi_\pi \pi_t^{CPI} + \phi_X \widehat{X}_t \right). \quad (195)$$

The ex ante real rate is

$$\widehat{R}_t = \widehat{i}_t - \mathbb{E}_t \pi_{H,t+1}. \quad (196)$$

UIP, including the optional NFA-premium robustness, is

$$\widehat{R}_t - \widehat{R}_t^* = \mathbb{E}_t (\widehat{q}_{t+1} - \widehat{q}_t) - \phi_{NFA} dNFA_t. \quad (197)$$

Here $dNFA_t$ is a level deviation normalized by $\bar{X}$; the baseline sets $\phi_{NFA} = 0$.

The labor-tax rule is

$$\bar{\tau}^\ell \widehat{\tau}_t^\ell = \psi_B \frac{B_{t-1} - \bar{B}}{\bar{X}}. \quad (198)$$

The government budget and mutual-fund balance sheet are linearized in level deviations. This convention is essential for objects such as NFA_t and TB_t , whose stationary values are zero; government debt is instead expanded around its positive stationary level $\bar{B}/\bar{X} = 2.32$.

B.17 Sequence-space solution

The household block maps sequences of aggregate inputs into sequences of household aggregates. Let

$$\mathbf{u} = \left\{ r_t^a, w_t, N_t, \tau_t^\ell, T_t, p_{M,t}^C \right\}_{t=0}^{T-1} \quad (199)$$

denote household inputs and let

$$\mathbf{h} = \{ A_t, C_t, S_t, C_{H,t}, C_{M,t}, \Lambda_t^e \}_{t=0}^{T-1} \quad (200)$$

denote household outputs.

Around the stationary equilibrium,

$$d\mathbf{h} = \mathbf{J}^H d\mathbf{u}, \quad (201)$$

where $\mathbf{J}^H$ contains the sequence-space Jacobians of the household block.

The household inputs themselves depend on endogenous aggregates and exogenous shocks. Writing

$$d\mathbf{u} = \mathbf{U}_y d\mathbf{y} + \mathbf{U}_z d\mathbf{z}, \quad \mathbf{J}_y^H \equiv \mathbf{J}^H \mathbf{U}_y, \quad \mathbf{J}_z^H \equiv \mathbf{J}^H \mathbf{U}_z, \quad (202)$$

defines the household derivatives with respect to the aggregate unknowns and the exogenous shocks used below.

The remaining equilibrium conditions can be stacked as

$$\mathcal{F}(\mathbf{y}, \mathbf{h}, \mathbf{z}) = 0, \quad (203)$$

where $\mathbf{y}$ collects endogenous aggregate sequences and $\mathbf{z}$ collects exogenous shocks.

Substituting equation (201) into the linearization of equation (203) gives

$$[\mathcal{F}_y + \mathcal{F}_h \mathbf{J}_y^H] d\mathbf{y} = -[\mathcal{F}_z + \mathcal{F}_h \mathbf{J}_z^H] d\mathbf{z}. \quad (204)$$

The equilibrium transition is

$$d\mathbf{y} = -[\mathcal{F}_y + \mathcal{F}_h \mathbf{J}_y^H]^{-1} [\mathcal{F}_z + \mathcal{F}_h \mathbf{J}_z^H] d\mathbf{z}. \quad (205)$$

The first-order transition is solved on $t = 0, \dots, T - 1$, with $T = 140$ quarters. Predetermined level deviations, including $dNFA_{-1}$, are initialized at zero. The forward-shift operator sets the missing date- T deviations in forward-looking equilibrium equations to their stationary values. No separate terminal restriction is imposed on NFA , which evolves recursively from the current account. Re-solving the system at $T = 100$ changes the first 24 quarters of output, composite consumption, imports, and CPI inflation by at most 0.0025, 0.0104, 0.0062, and 0.0021 percentage points, respectively. This is a short-run horizon check; it does not imply that the terminal NFA stock is zero.

For the tariff experiment, $\mathbf{J}_z^H d\mathbf{z}$ contains the household block's direct response to the final-import price path, whereas $\mathcal{F}_z d\mathbf{z}$ contains direct entries outside that block, including tariff revenue, external-price conditions, and imported-input costs. This allocation is a bookkeeping convention: Their sum determines the same equilibrium response. The resolvent in equation (205) propagates those direct entries through wages, employment, returns, prices, policy, fiscal debt, and net foreign assets. It does not turn the fixed-basket sufficient statistic into a decomposition of domestic value added.

Separate household Jacobians are computed for the common-basket, vertical-exposure, and full-exposure economies. The general-equilibrium blocks are otherwise held fixed.

C External validation using Spanish expenditure data

This appendix evaluates the portability of the benchmark exposure structure using Spanish data; it does not calibrate the model to Spain. I combine the 2022 Spanish Household Budget Survey (EPF) with the 2022 INE input–output tables to construct an external validation of direct-final import exposure. The analytical sample contains 20,585 households with positive total expenditure and finite imputed exposure. Household population weights use the factors revised to the 2021 Census. The original EPF factor is used to recover household-level total and item expenditure from the corresponding expanded variables.

Let x_{ig} denote household i ’s expenditure in EPF category g , let X_i denote total EPF expenditure, and let d_g denote the category’s imputed direct-final import intensity. Household exposure is

$$\hat{\alpha}_i = \frac{\sum_g x_{ig} d_g}{X_i}. \quad (206)$$

The intensities are constructed at CPA64 level as imported household final use divided by total household final use. EPF COICOP categories are assigned to CPA64 industries through the many-to-many concordance. In the benchmark mapping, multiple CPA candidates are allocated using INE household-consumption weights; conservative and alternative plausible mappings are reported as sensitivity checks. Direct-final imports are kept separate from total imported content, which additionally includes imported intermediate inputs and is not a measure of the direct consumer-price exposure in equation (206).

I form ten expenditure groups using population-weighted quantiles of X_i . For decile q , define

$$\bar{\alpha}_q = \frac{\sum_{i \in q} w_i \hat{\alpha}_i}{\sum_{i \in q} w_i}, \quad \hat{\sigma}_q = \left(\frac{\sum_{i \in q} w_i (\hat{\alpha}_i - \bar{\alpha}_q)^2}{\sum_{i \in q} w_i} \right)^{1/2}. \quad (207)$$

The model-aligned horizontal-dispersion statistic is the unweighted cell average

$$\hat{H}_{\text{EPF}} = \frac{1}{10} \sum_{q=1}^{10} \hat{\sigma}_q. \quad (208)$$

This construction yields $\hat{H}_{\text{EPF}} = 0.0959049$. By contrast, an alternative expenditure-weighted aggregation yields 0.1075764: it weights observations within each decile by $w_i X_i$ and aggregates decile standard deviations using expenditure shares. It is not definitionally aligned with equation (208).

Table 5: Impact distributions in the vertical and full economies

Node	Outcome	Mean	Standard deviation	$p90 - p10$
Benchmark	Cost of living	2.2920/2.2923/ + 0.0003	0.2672/0.3118/ + 0.0446	0.5402/0.7399/ + 0.1996
	Expenditure	-2.1678/ - 2.1682/ - 0.0004	1.3961/1.3966/ + 0.0005	3.8856/3.8869/ + 0.0013
	Composite consumption	-5.2466/ - 5.2470/ - 0.0005	2.5172/2.5236/ + 0.0064	5.8153/5.8470/ + 0.0317
Spanish-EPF central	Cost of living	2.2920/2.2976/ + 0.0056	0.2672/0.7518/ + 0.4846	0.5402/2.0768/ + 1.5366
	Expenditure	-2.1678/ - 2.1747/ - 0.0069	1.3961/1.4055/ + 0.0094	3.8856/3.9102/ + 0.0246
	Composite consumption	-5.2466/ - 5.2554/ - 0.0088	2.5172/2.6358/ + 0.1187	5.8153/6.4083/ + 0.5930

Notes: Each entry reports vertical / full / (full minus vertical). Quantities are percentage points in the cross-sectional distribution of date-0 household responses. Cost of living is the compensating-expenditure change for the stationary utility bundle; expenditure and composite consumption are percentage responses. The Spanish-EPF central node matches $\hat{H}_{EPF} = 0.0959049$ while preserving mean ω ; its basket-dispersion coefficient is 0.644, or 4.58 times the benchmark value. It is an empirical sensitivity, not a structural calibration. These objects are not welfare measures. Source: Model calculations.

Table 4: EPF validation of within-group import-exposure dispersion

Definition or sensitivity	Estimate	Interpretation
Expenditure-weighted aggregation	0.1075764	Alternative weighting
Population-weighted within-decile statistic	0.0959049	Main external-validation estimate
Conservative COICOP-CPA mapping	0.0949852	Mapping sensitivity
Alternative plausible mapping	0.0966726	Mapping sensitivity
Winsorization, 1st-99th percentiles	0.0940362	Outlier sensitivity
Winsorization, 2.5th-97.5th percentiles	0.0902811	Outlier sensitivity
Spanish-EPF low node	0.0943163	Pseudo-bootstrap percentile
Spanish-EPF central node	0.0959049	Main estimate
Spanish-EPF high node	0.0976294	Pseudo-bootstrap percentile

Notes: The model-aligned statistic forms population-weighted expenditure deciles, computes population-weighted standard deviations within deciles, and averages the ten cell standard deviations equally. Mapping variants hold that formula fixed. Winsorization is applied before recomputing the statistic. The low and high nodes are the 2.5th and 97.5th percentiles of a 500-draw household-level Poisson pseudo-bootstrap that recomputes deciles; they are approximate resampling sensitivities, not endpoints of a design-based confidence interval. Neither range identifies the covariance between exposure and household iMPCs.

Categories assigned zero import intensity and unassigned expenditure remain in the denominator and contribute zero to the numerator. Import intensities are restricted to the unit interval. The baseline does not winsorize exposure. Households without positive total expenditure are excluded. Because the public files do not support complete PSU-, stratum-, or replicate-weight inference, the pseudo-bootstrap is reported only as approximate resampling uncertainty; mapping sensitivity is likewise separate from sampling uncertainty.

Several limitations remain. EPF records expenditure rather than physical quantities or super-numerary consumption, so imputed exposure can mix prices, product quality, infrequent purchases, and concordance error. CPA64 aggregation also suppresses within-industry variation in import origin. The exercise disciplines the marginal dispersion of exposure; it does not observe household iMPCs and does not identify the exposure-iMPC covariance.

Table 5 and Figure 7 isolate the incremental effect of permanent basket types. Each comparison holds fixed the vertical expenditure gradient and uses common fractional expenditure groups.

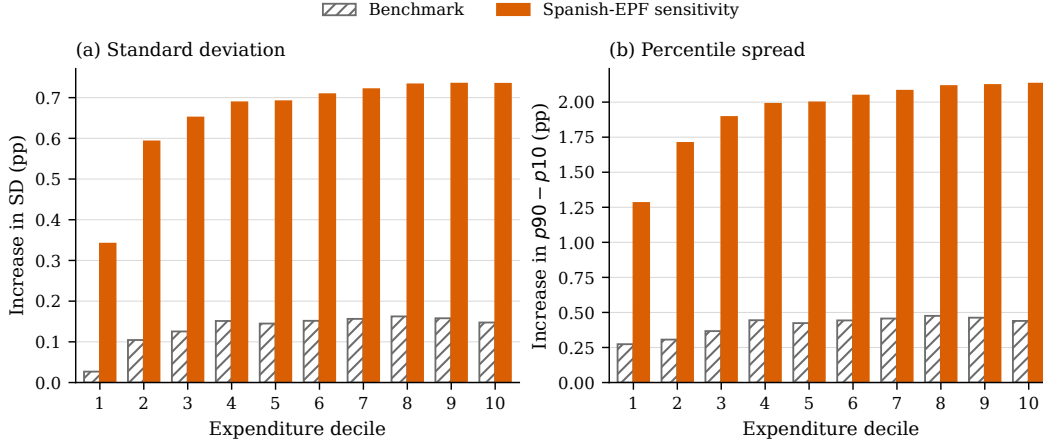


Figure 7: Incremental cost-of-living dispersion. Full-minus-vertical differences in the dispersion of the impact cost-of-living change, by fixed expenditure decile. The benchmark uses the basket-dispersion parameter reported in Table 1. The Spanish-EPF central sensitivity matches $\hat{H}_{\text{EPF}} = 0.0959049$ while preserving mean ω ; it is not a structural calibration. Cost of living is a compensating-expenditure object for the stationary utility bundle, not a welfare measure. Source: Model calculations.

D Robustness

This appendix examines the robustness of the main results along four dimensions. First, it separates tariffs on final consumption goods from tariffs on imported inputs. Second, it varies the tariff process, monetary policy, nominal rigidity, production, household demand, the external risk premium, and a uniform-rebate fiscal closure. Third, it allows intratemporal basket composition to adjust sluggishly. Finally, it verifies that the results are stable to the resolution of the household asset grid.

Two conclusions are robust. The level of the aggregate response depends on the production structure, policy closure, and speed of basket adjustment. Heterogeneous exposure makes the impact output response more negative than in the common-basket economy in every specification considered, but its magnitude is not invariant: sluggish basket adjustment compresses the full-common gap substantially. Permanent basket types have a negligible incremental aggregate effect throughout.

D.1 Final-goods and imported-input tariffs

Figure 8 decomposes the difference between the full-exposure and common-basket economies by tariff margin.

The household channel. A tariff on final consumption imports makes impact domestic value added approximately 0.066 percentage points lower in the full-exposure economy. At the same time, composite consumption falls around 0.412 percentage points less and total import quantities fall around 0.812 percentage points less.

These responses reflect weaker expenditure switching. Necessity-like import demand makes

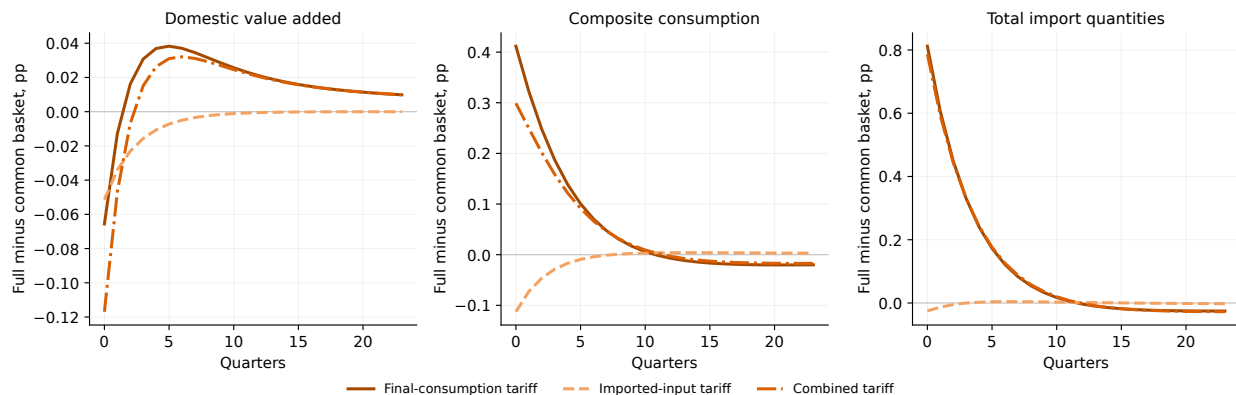


Figure 8: Exposure gaps across tariff margins. The figure reports the full-exposure response minus the common-basket response under a tariff on final consumption imports, a tariff on imported inputs, and a common tariff on both uses. Differences are expressed in percentage points.

highly exposed households reduce imported consumption less aggressively. They preserve more imports but redirect less expenditure toward domestic goods. Domestic value added therefore falls more even though aggregate composite consumption falls less.

The production channel. A tariff on imported inputs makes impact domestic value added approximately 0.051 percentage points lower in the full-exposure economy. It also makes composite consumption slightly more contractionary. This margin does not directly change household import prices, but it affects households through labor income, asset returns, and the equilibrium response of domestic production.

Because the model is solved to first order, the response to the common tariff is the sum of the two separate experiments. Both channels contribute to the additional fall in domestic value added. Most of the difference in household consumption and import quantities, however, comes from the final-consumption tariff.

The gaps are concentrated in the first few quarters. Exposure heterogeneity changes the impact and early transition but has little effect once the tariff wedge dissipates.

D.2 Alternative calibrations and policy closures

Table 6 reports the corresponding impact responses and the output difference relative to the common-basket economy.

Tariff persistence and monetary policy. A more persistent tariff produces a larger output contraction, while a less persistent tariff attenuates it. A stronger CPI response also deepens the fall in activity. These parameters materially change the level of the aggregate response but have little effect on the full-common difference.

The same pattern holds for wage rigidity. Greater wage rigidity increases the contraction and modestly enlarges the exposure gap. More flexible wages reduce both.

Table 6: Alternative calibrations and policy closures

Specification	Output	Consumption	Imports	CPI inflation	Full- common output
<i>Panel A: Baseline, tariff process, and nominal block</i>					
Baseline	-1.56	-4.93	-7.63	4.17	-0.117
Tariff persistence: $\rho_\tau = 0.50$	-1.33	-4.78	-7.58	4.37	-0.127
Tariff persistence: $\rho_\tau = 0.90$	-2.02	-5.23	-7.70	3.70	-0.102
Taylor coefficient: $\phi_\pi = 1.10$	-1.12	-4.64	-7.55	4.52	-0.121
Taylor coefficient: $\phi_\pi = 2.00$	-1.99	-5.22	-7.70	3.85	-0.116
Wage Phillips slope: $\kappa_W = 0.025$	-1.65	-5.00	-7.64	4.23	-0.138
Wage Phillips slope: $\kappa_W = 0.100$	-1.41	-4.84	-7.60	4.09	-0.090
<i>Panel B: Production, external, and fiscal blocks</i>					
Input elasticity: $\nu_I = 0.25$	-2.03	-5.05	-6.68	4.14	-0.122
Input elasticity: $\nu_I = 1.00$	-0.66	-4.74	-9.44	4.24	-0.108
NFA premium: $\phi_{NFA} = 0.02$	-2.06	-4.06	-7.03	3.51	-0.055
Uniform revenue rebate	-1.54	-3.34	-6.72	4.33	-0.080
<i>Panel C: Household demand</i>					
Final-good elasticity: $\eta_C = 1.00$	-1.82	-5.00	-6.96	4.15	-0.080
Final-good elasticity: $\eta_C = 2.00$	-1.31	-4.88	-8.28	4.19	-0.153

Notes: Output, consumption, and total imports are impact percentage deviations in the full-exposure economy. CPI inflation and the full-common output difference are in percentage points. The baseline tariff is applied jointly to final consumption goods and imported inputs.

Production and external adjustment. The elasticity between domestic value added and imported inputs is important for the level of output. When inputs are difficult to replace, the tariff generates a larger domestic contraction. With a unit elasticity, firms substitute more strongly and the output loss becomes substantially smaller, although imports fall more.

Introducing a debt-elastic NFA premium increases the output contraction and reduces the full-common gap. External adjustment therefore affects the quantitative size of the household mechanism, but not its sign.

Fiscal closure. A uniform rebate substantially cushions household consumption but leaves the impact output contraction close to the baseline. Relative to the baseline, the full-common output gap narrows from -0.117 to -0.080 percentage points. Revenue recycling therefore affects incidence and the quantitative size of the household channel without reversing the contractionary output response in the implemented closures.

Household substitution. Changing the elasticity between domestic and imported final goods alters both expenditure switching and the aggregate contraction. A lower elasticity produces a larger output decline but a smaller full-common gap. A higher elasticity produces a smaller aggregate contraction but a larger exposure gap.

Across all frictionless-basket specifications in Table 6, the full-exposure impact output response is between 0.055 and 0.153 percentage points more negative than in the common-basket economy. The result is therefore not driven by a single calibration or policy rule within the benchmark demand system. The next exercise varies the timing of intratemporal basket adjustment itself.

D.3 Sluggish intratemporal basket adjustment

The benchmark Stone–Geary block adjusts the supernumerary domestic/import composition frictionlessly. Let

$$b_{i,t} \equiv d \log \left(\frac{\tilde{c}_{M,i,t}}{\tilde{c}_{H,i,t}} \right) \quad (209)$$

denote the log deviation of permanent basket component i 's actual supernumerary import-to-domestic quantity ratio, and let $b_{i,t}^*$ denote its frictionless Stone–Geary target. I use the local linear–quadratic tracking problem

$$\min_{\{b_{i,t}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \bar{\beta}_b^t \left[\frac{1}{2} (b_{i,t} - b_{i,t}^*)^2 + \frac{\lambda_b}{2} (b_{i,t} - b_{i,t-1})^2 \right], \quad b_{i,-1} = 0. \quad (210)$$

This is a local reduced-form sensitivity for composition timing, not a separately estimated structural household friction. Its first-order condition is

$$[1 + \lambda_b(1 + \bar{\beta}_b)]b_{i,t} - \lambda_b b_{i,t-1} - \bar{\beta}_b \lambda_b \mathbb{E}_t b_{i,t+1} = b_{i,t}^*. \quad (211)$$

Parameterizing the stable root by $\rho_b \in [0, 1)$ gives

$$\lambda_b = \frac{\rho_b}{(1 - \rho_b)(1 - \bar{\beta}_b \rho_b)}, \quad 0 < \bar{\beta}_b < 1. \quad (212)$$

For $0 < \rho_b < 1$, the other root is $1/(\bar{\beta}_b \rho_b) > 1$. The case $\rho_b = 0$ is treated separately below. In the finite-horizon implementation, the predetermined state is $b_{i,-1} = 0$ and the terminal condition is $\mathbb{E}_{T-1} b_{i,T} = 0$.

Let stationary supernumerary quantities be $\tilde{c}_{H,i}$ and $\tilde{c}_{M,i}$. The filtered ratio is mapped into quantities through

$$\begin{aligned} d\tilde{c}_{H,i,t} &= d\tilde{c}_{H,i,t}^* - \chi_i(b_{i,t} - b_{i,t}^*), \\ d\tilde{c}_{M,i,t} &= d\tilde{c}_{M,i,t}^* + \chi_i(b_{i,t} - b_{i,t}^*), \\ \chi_i &\equiv \frac{\tilde{c}_{H,i}\tilde{c}_{M,i}}{\tilde{c}_{H,i} + \tilde{c}_{M,i}}. \end{aligned} \quad (213)$$

At $\bar{p}_M^C = 1$, this is a zero-sum first-order reallocation. It preserves the baseline Jacobians for total expenditure, saving, composite consumption, and marginal utility; these are imposed unchanged rather than re-solved. This preservation is local because composition misallocation and the LQ loss are second order at the stationary allocation. The transformation is applied to each permanent basket component before aggregation, after which general equilibrium is resolved separately for each exposure economy.

The case $\rho_b = 0$ implies $\lambda_b = 0$ and reproduces frictionless adjustment exactly. As $\rho_b \uparrow 1$, $\lambda_b \rightarrow \infty$ and the supernumerary composition remains fixed over finite horizons. This fixes the composition

ratio, not total expenditure or quantities. The block requires positive stationary domestic and imported supernumerary quantities and does not guarantee positivity under arbitrarily large scaled first-order paths.

Table 7 reports a transparent persistence grid. The positive nodes are informed by descriptive CEX 2007–2009 basket-share persistence exercises, but CEX does not identify the structural adjustment parameter. They are therefore sensitivity nodes, not estimates. For each positive persistence node, brackets span discounting values $\bar{\beta}_b \in \{0.95, 0.98, 0.99\}$.

Table 7: Sluggish basket-adjustment sensitivity

Basket adjustment	Full output	Full composite consumption	Full final imports	Full-common output	Full-vertical output
Frictionless ($\rho_b = 0$)	−1.563	−4.930	−9.266	−0.117	−0.0009
Fast descriptive (0.414)	[−1.982, −1.979]	[−5.001, −5.000]	[−6.532, −6.508]	[−0.057, −0.057]	[0.0006, 0.0006]
Intermediate (0.695)	[−2.225, −2.220]	[−5.053, −5.050]	[−4.829, −4.784]	[−0.022, −0.022]	[0.0014, 0.0015]
Latent alternative (0.815)	[−2.303, −2.299]	[−5.079, −5.074]	[−4.207, −4.157]	[−0.011, −0.010]	[0.0017, 0.0017]
Slow (0.920)	[−2.346, −2.344]	[−5.106, −5.099]	[−3.781, −3.743]	[−0.0043, −0.0040]	[0.0019, 0.0019]
Fixed-composition limit	−2.353	−5.128	−3.606	−0.0027	0.0019

Notes: Impact responses to the combined tariff shock. Levels are percentage deviations; output gaps are percentage points. Brackets span $\bar{\beta}_b \in \{0.95, 0.98, 0.99\}$. Here ρ_b is the stable root and $\bar{\beta}_b$ is an auxiliary sensitivity parameter. Positive persistence nodes are CEX-informed sensitivities, not structural estimates. The final row is the fixed-supernumerary-ratio limit of this block, not a universal theoretical bound. The underlying model remains first-order.

Sluggish adjustment prevents the immediate expenditure switch from imports toward the domestic good. It therefore makes the aggregate recession substantially larger: full-exposure impact output falls by 2.35% in the fixed-composition limit, compared with 1.56% in the benchmark. At the same time, the full-common output gap shrinks from −0.117 to −0.0027 percentage points. Thus the level of the recession and the incremental role of exposure heterogeneity move in opposite directions. The already tiny full-vertical difference also changes sign at positive persistence and remains economically negligible throughout. The robust conclusion is that permanent basket-type dispersion has little additional aggregate effect; the magnitude, though not the sign, of the full-common gap is conditional on rapid basket adjustment.

D.4 Numerical accuracy

Table 8 recomputes the stationary household block and the fixed-basket direct-statistic decomposition using alternative asset-grid resolutions.

Table 8: Asset-grid robustness

Grid points	Assets/GDP	Quarterly MPC	Four-quarter MPC	Import share	Direct	Exposure-iMPC
120	10.0826	0.19996	0.28262	0.29996	−0.42482	0.43924
160	10.0000	0.20000	0.28292	0.30000	−0.42526	0.43926
200	9.9529	0.20005	0.28318	0.30002	−0.42558	0.43933

Notes: The baseline uses 160 asset-grid points. The fixed-basket direct statistic is expressed as a percentage of aggregate household expenditure.

The quarterly MPC, aggregate import share, exposure–iMPC correlation, and fixed-basket direct statistic are essentially unchanged across grids. The small movement in aggregate assets reflects approximation at the upper part of the wealth distribution and does not affect the mechanism.

Across all general-equilibrium robustness exercises, the maximum residual of the stacked equilibrium system remains below 3×10^{-13} . The numerical implementation also preserves the direct-effect decomposition exactly and is stable to alternative equilibrium closures and solution horizons. This residual measures the accuracy with which the stated first-order system is solved; it is not a measure of the nonlinear approximation error at a 0.10 log-wedge shock. Exact proportional scaling with shock size is likewise mechanical in a linear model. Local nonlinear checks of the isolated household price block agree closely with its Jacobian for import-price changes between 10^{-4} and 10^{-2} , but the implementation does not solve a nonlinear general-equilibrium transition. Accordingly, all transition results are interpreted as local first-order experiments.